Biochemistry



Section N
Nucleotide Metabolism

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Biochemistry and Molecular Biology, College of Life Sciences

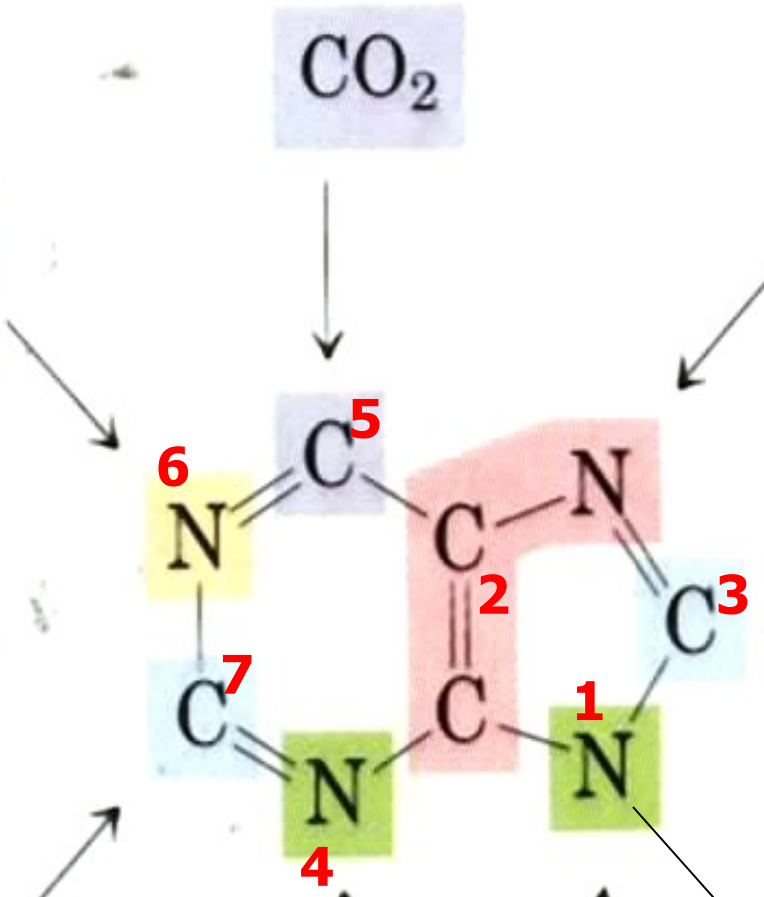
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13854808762

Radioisotope tracer experiments revealed the origins of the atoms in the purine and pyrimidine rings



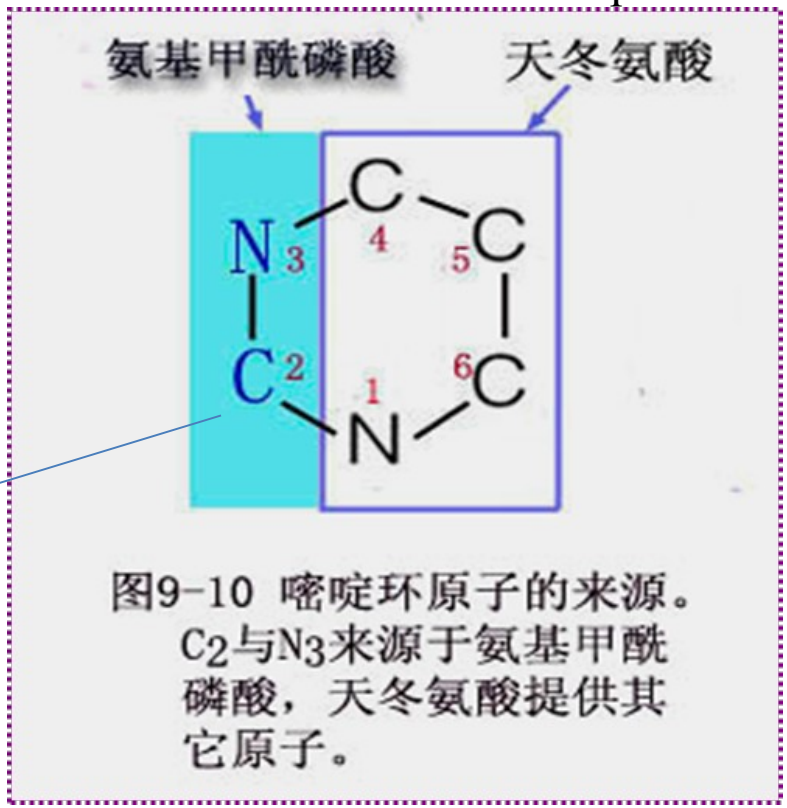
origins of the ring atoms of purines (*numbered is the order of addition*)



imidazole ring first

Carbamoyl phosphate

Aspartate

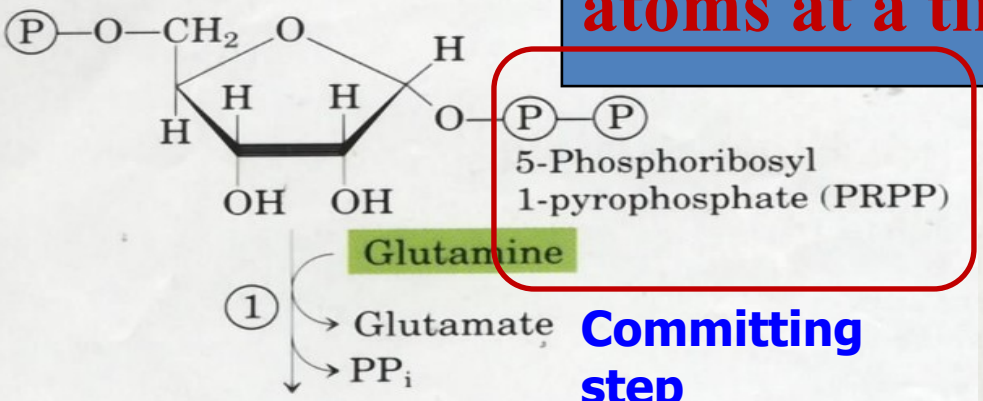


The atoms of the pyrimidine rings were revealed to be derived from HCO_3^-, Gln and Asp.

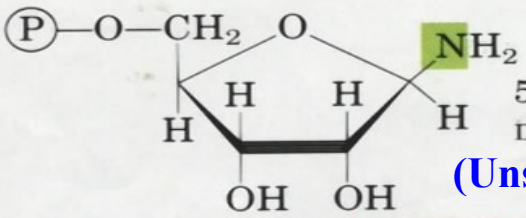
***De novo* purine nucleotide synthesis: the base assembles on the ribose phosphate; IMP is first nucleotide synthesized.**



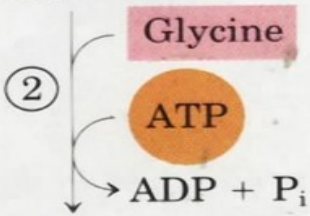
The purine ring is built up one or three atoms at a time on ribose 5-P



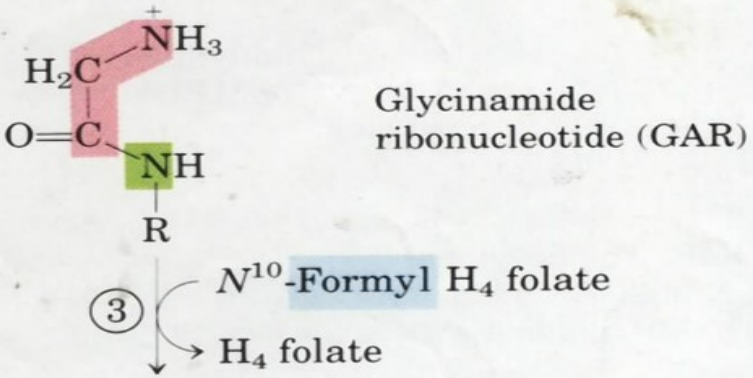
Committing step

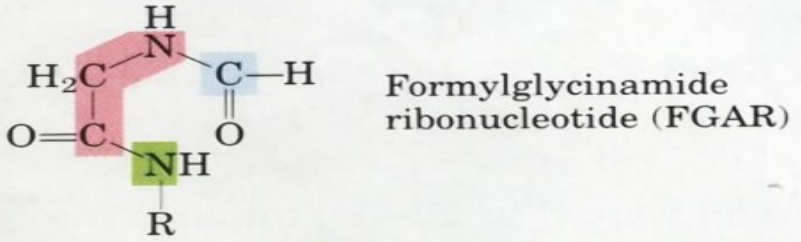


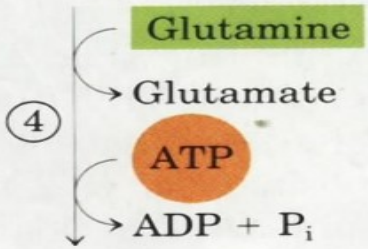
(Unstable)

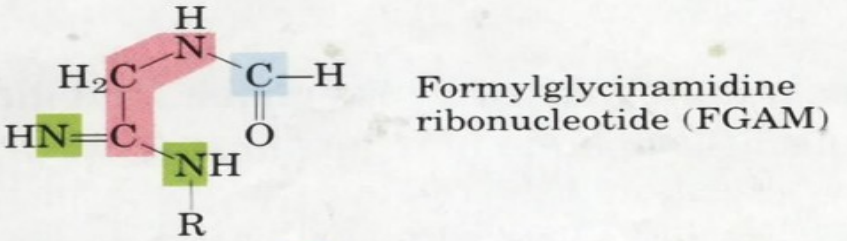


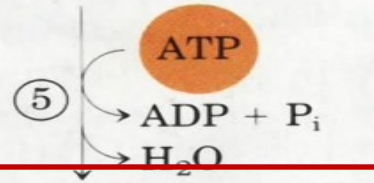
1, 3, 5 are catalyzed by one protein!

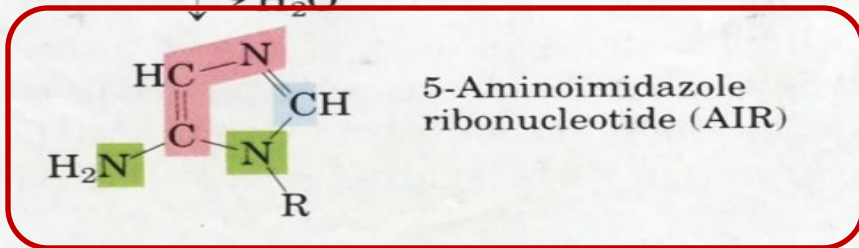


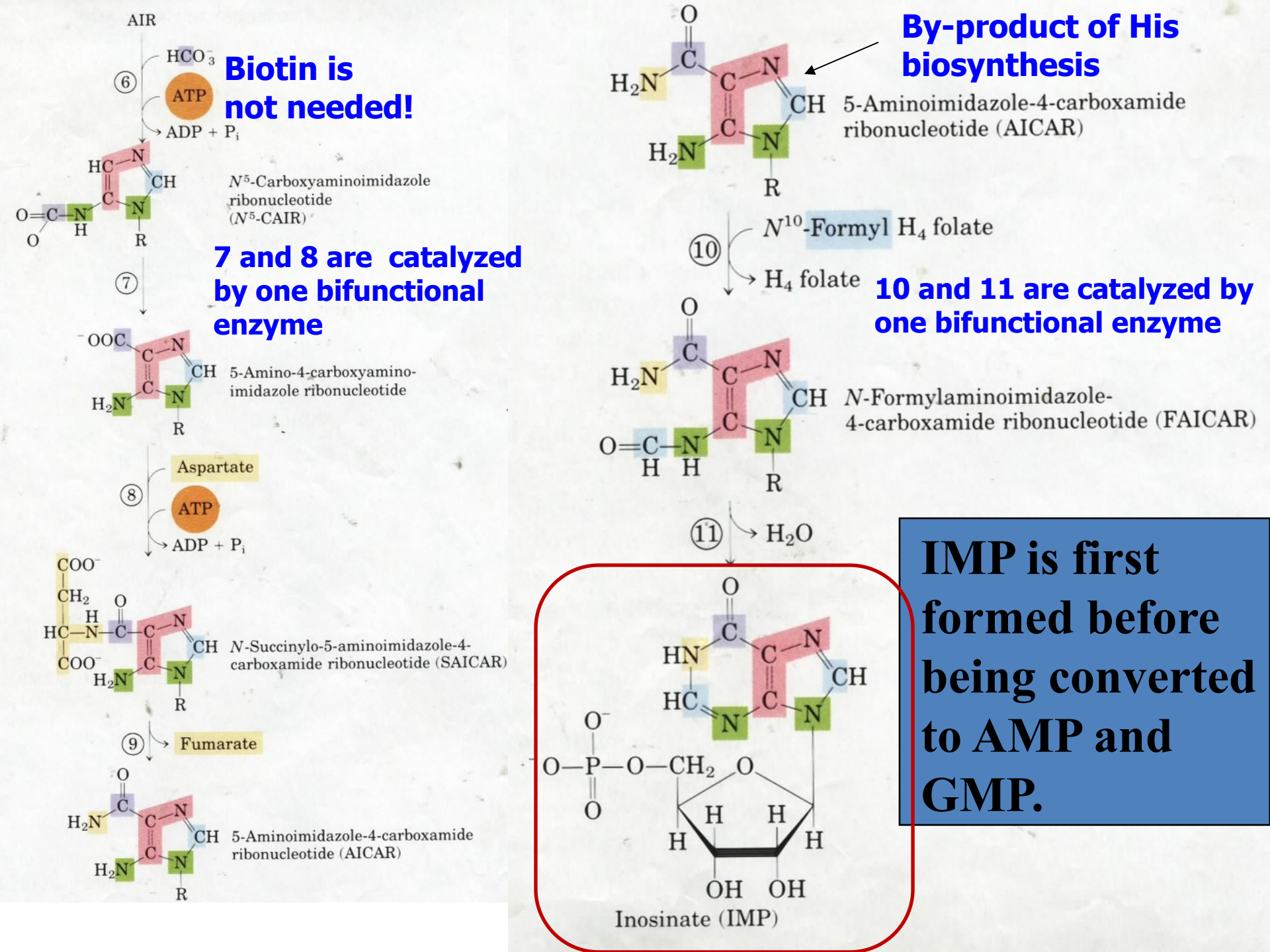












Biotin is not needed!

7 and 8 are catalyzed by one bifunctional enzyme

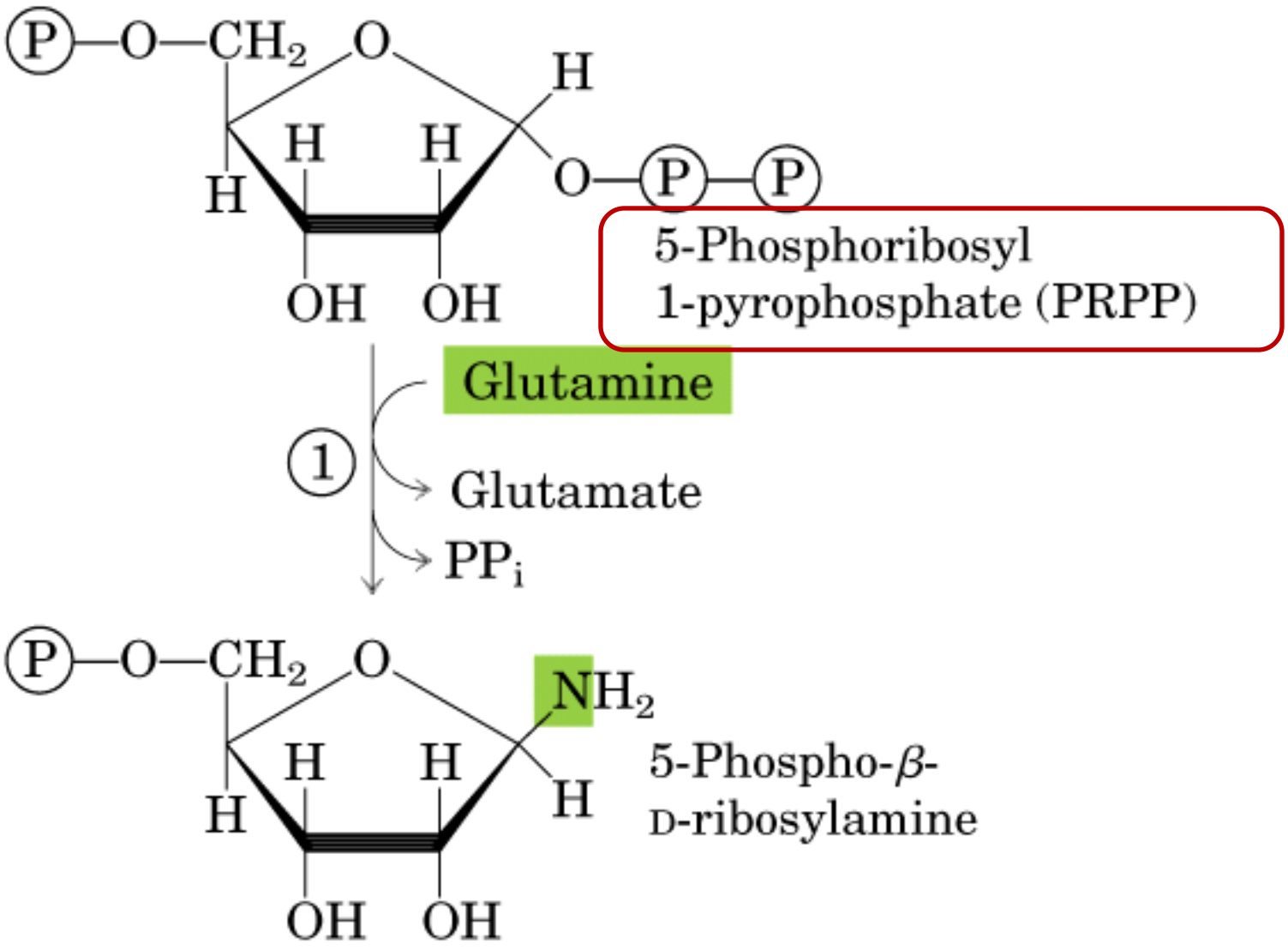
By-product of His biosynthesis

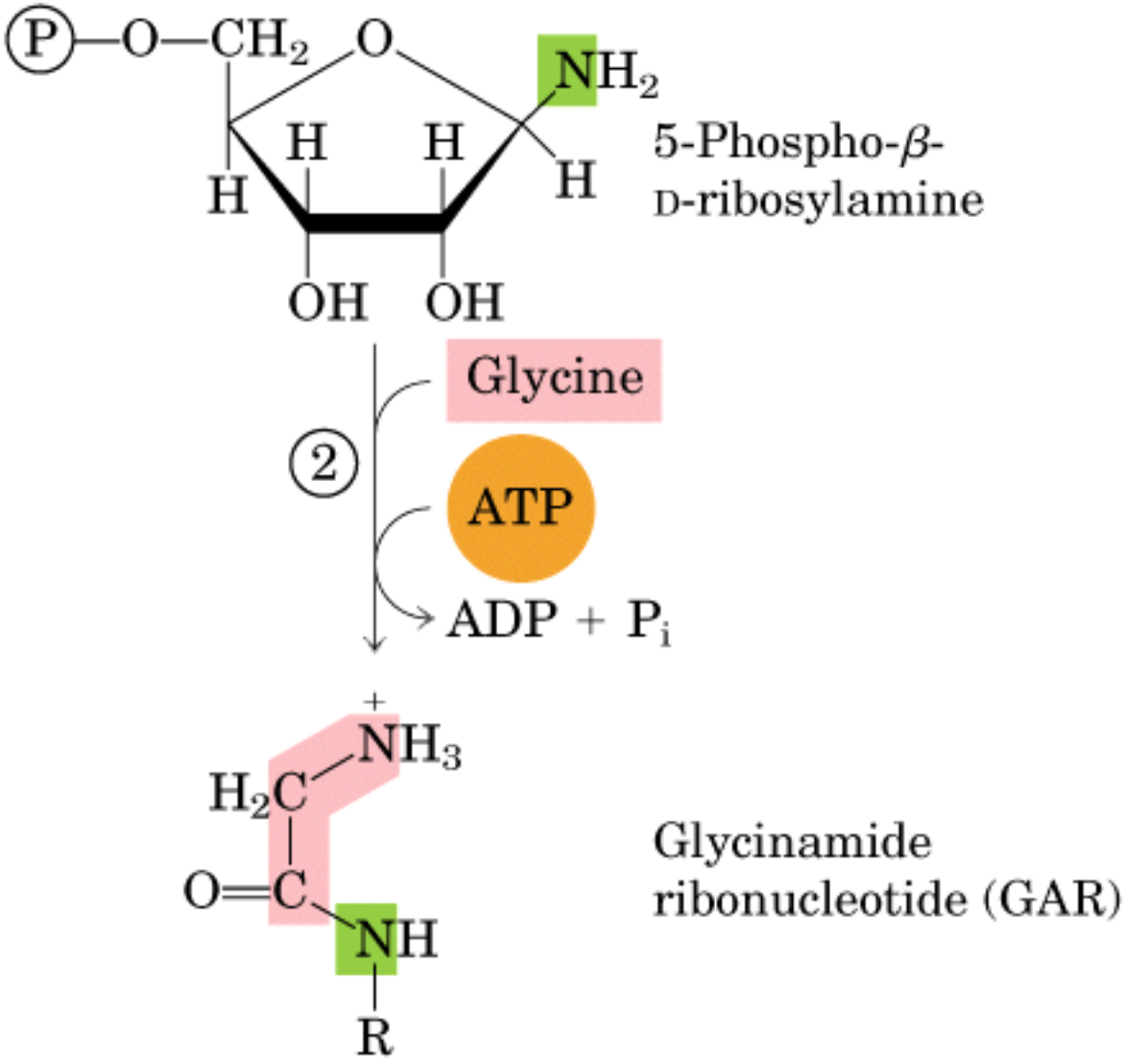
10 and 11 are catalyzed by one bifunctional enzyme

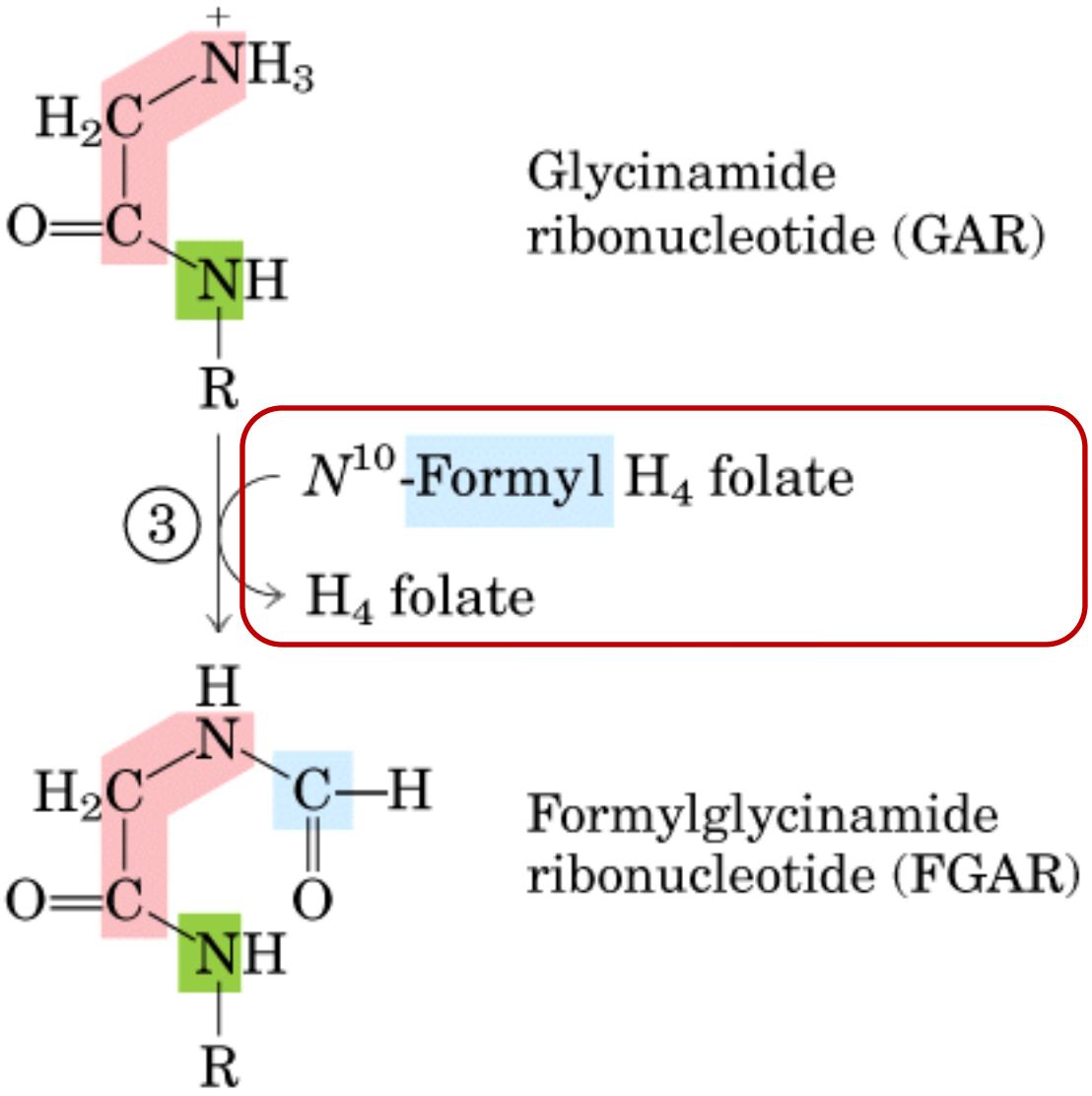
IMP is first formed before being converted to AMP and GMP.

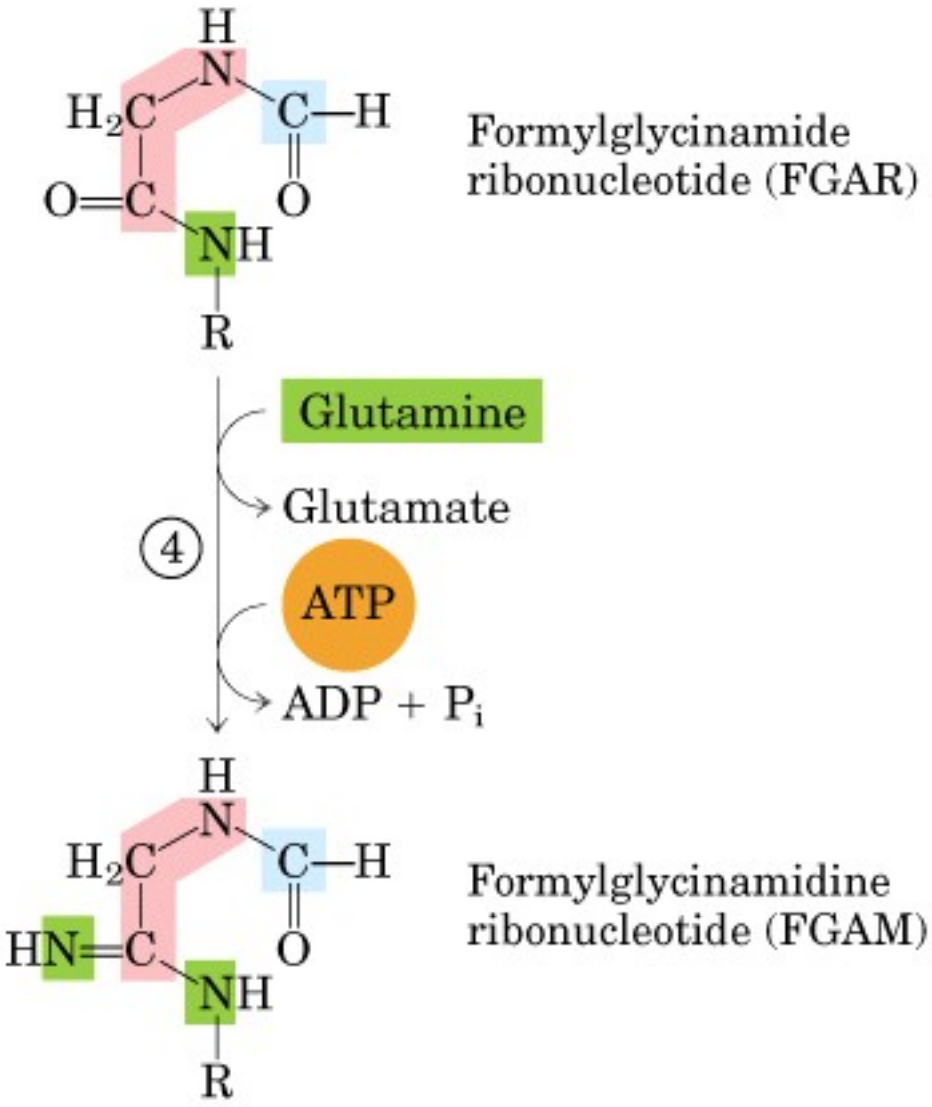
De novo purine nucleotide synthesis: AMP and GMP are derived from IMP.




P—O—CH2
O
H
H
H
H
O—P—P
OH
OH
5-Phosphoribosyl
1-pyrophosphate (PRPP)
Glutamine
1
Glutamate
PPi
P—O—CH2
O
NH2
H
H
H
H
OH
OH
5-Phospho-β-
D-ribosylamine

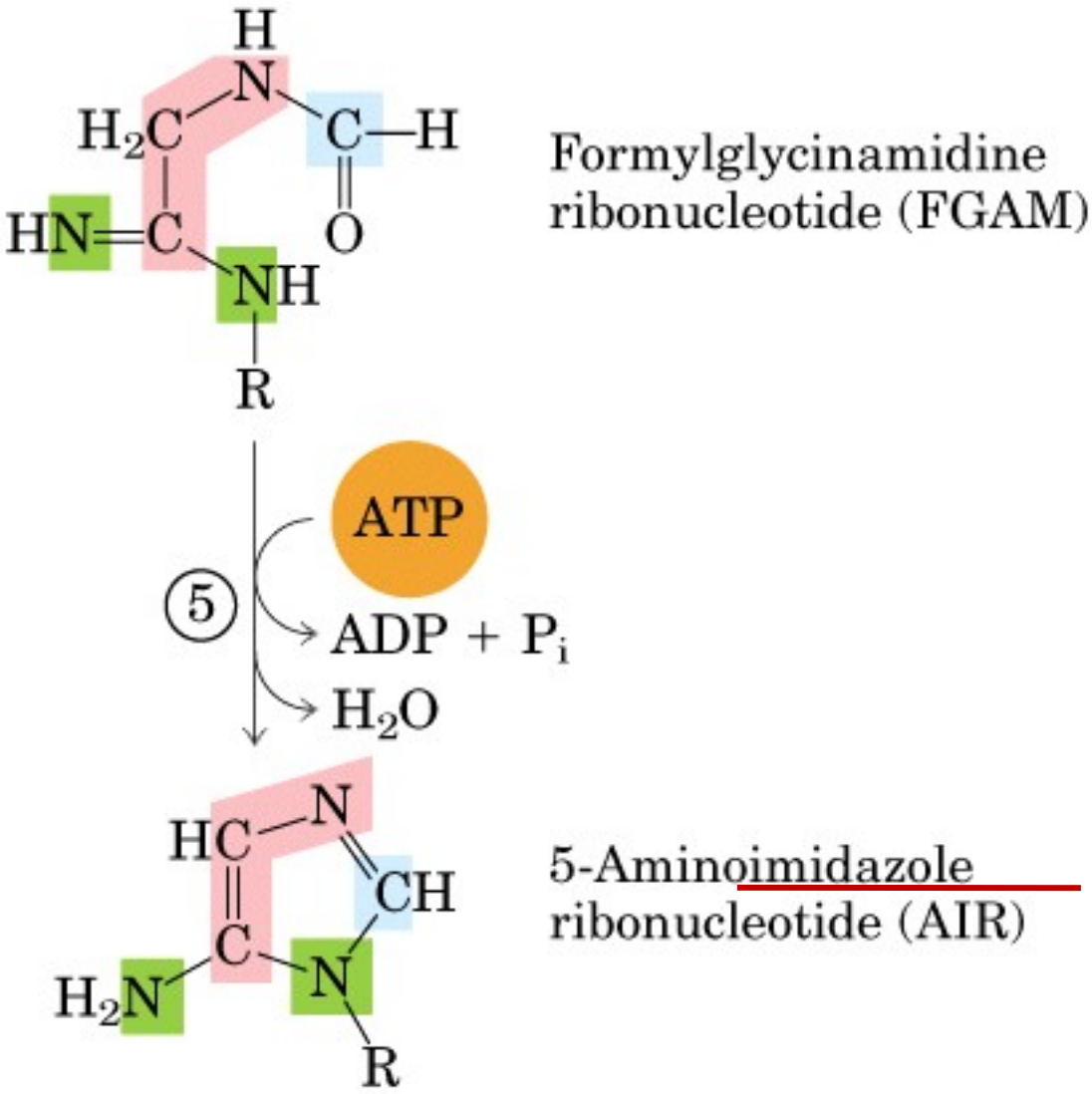

P—O—CH2
O
NH2
H
H
H
H
OH
OH
5-Phospho-β-
D-ribosylamine
Glycine
2
ATP
ADP + Pi
+
NH3
H2C
O=C
NH
R
Glycinamide
ribonucleotide (GAR)


Glycinamide ribonucleotide (GAR)
3
N^{10}-Formyl H_4 folate
H_4 folate
Formylglycinamide ribonucleotide (FGAR)



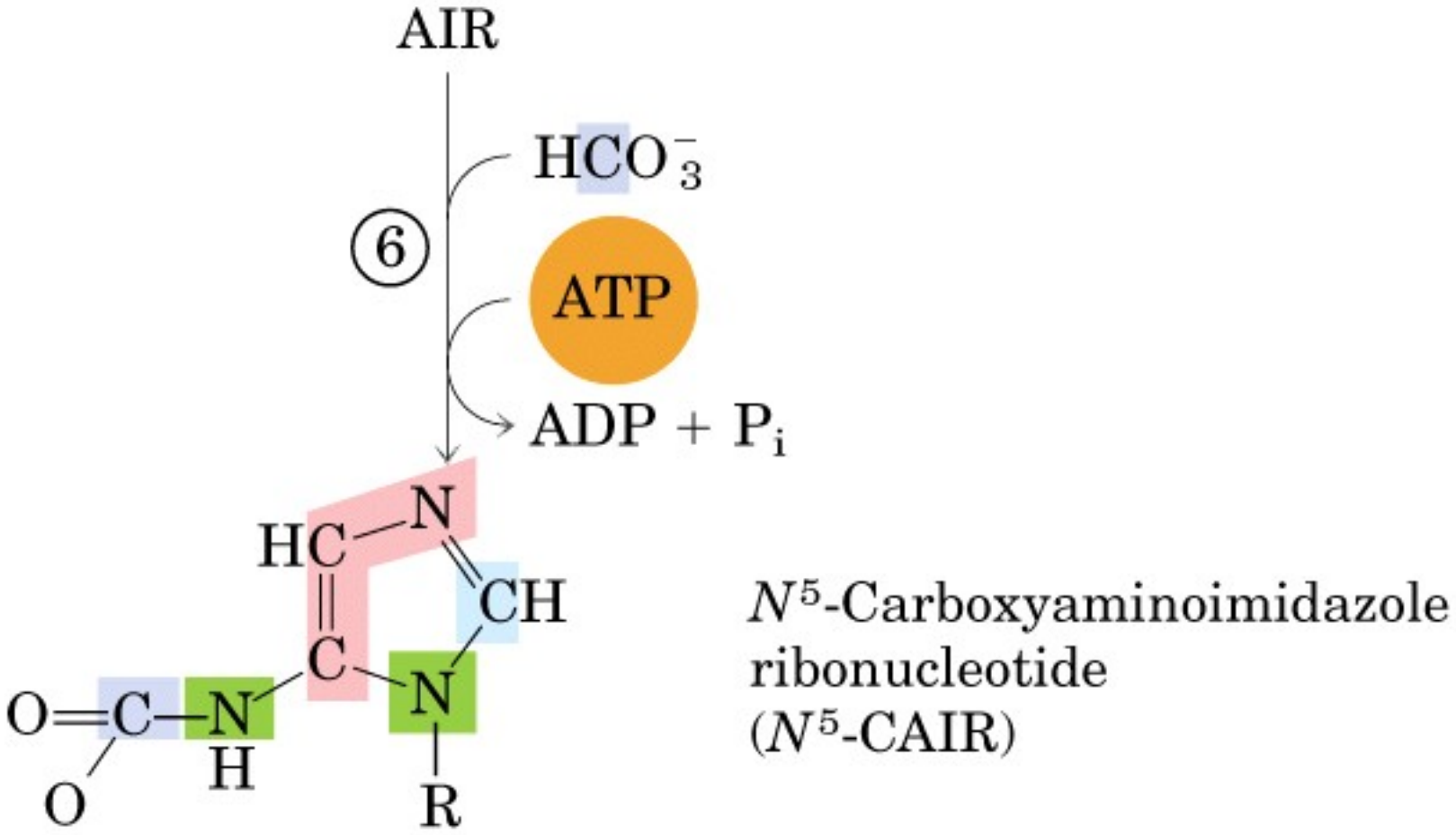
Formylglycinamide ribonucleotide (FGAR)

Formylglycinamidine ribonucleotide (FGAM)

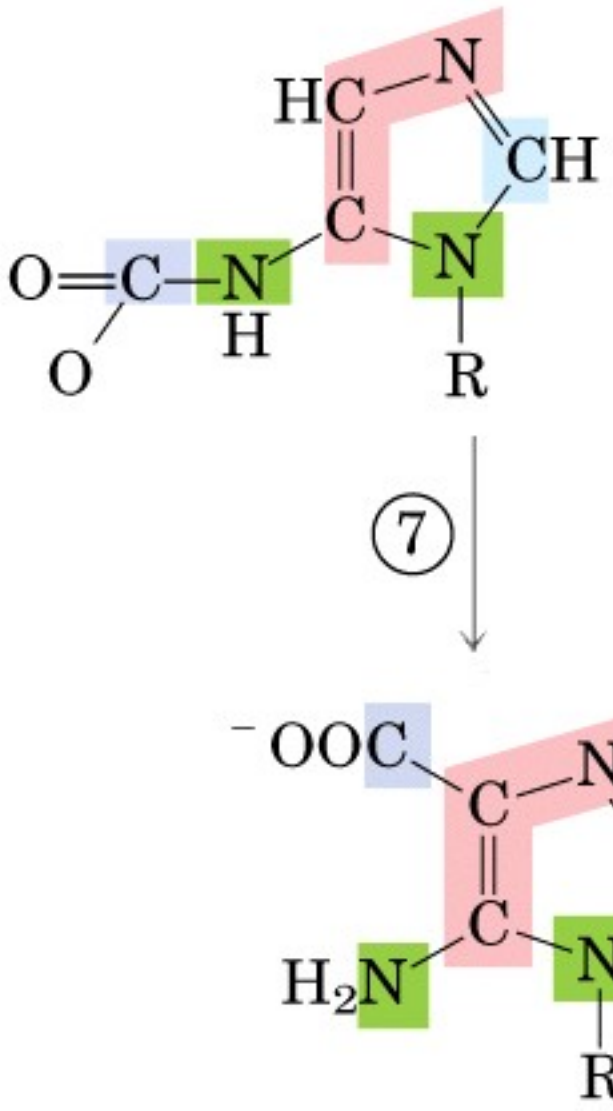


Formylglycinamidine ribonucleotide (FGAM)

5-Aminoimidazole ribonucleotide (AIR)

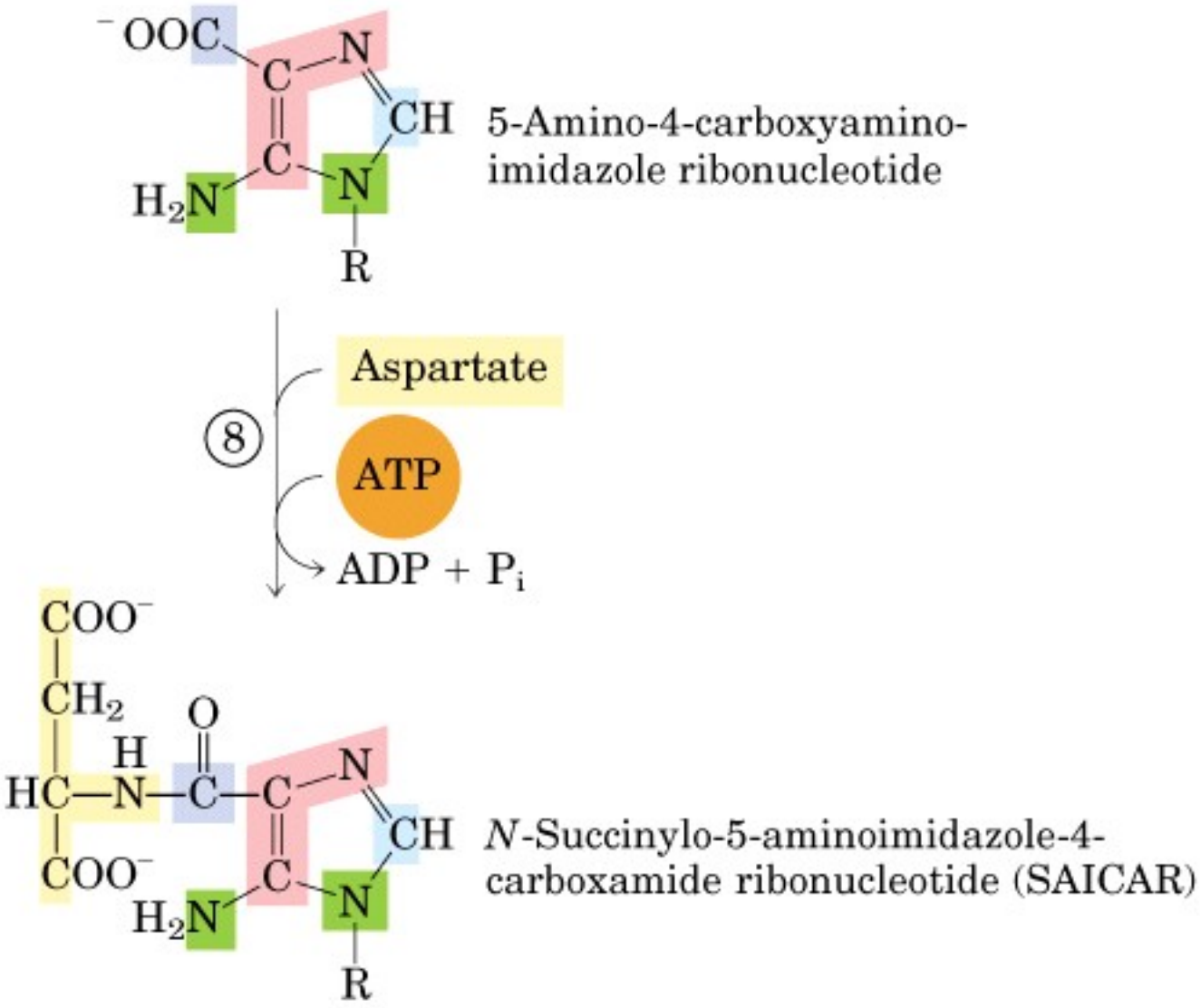


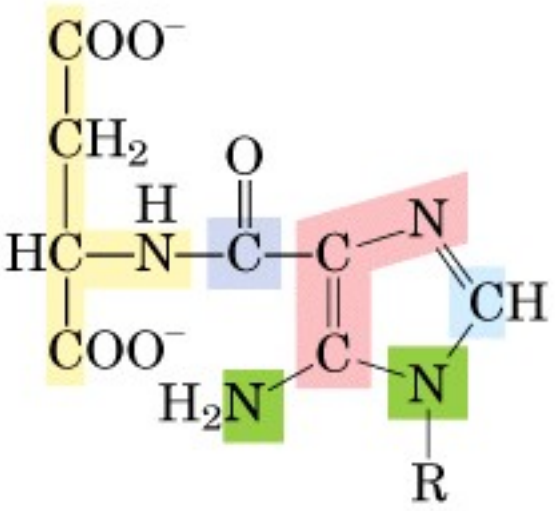
N^5-Carboxyaminoimidazole ribonucleotide (N^5-CAIR)



N^5-Carboxyaminoimidazole ribonucleotide (N^5-CAIR)

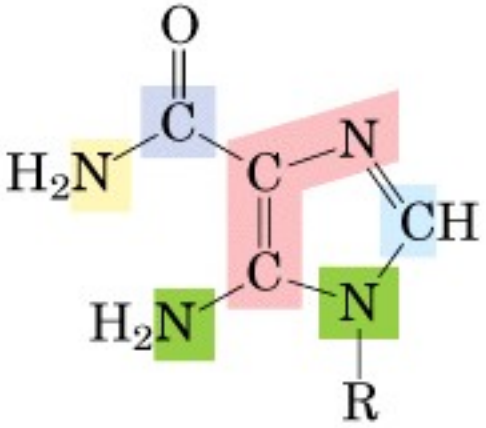
5-Amino-4-carboxyamino-imidazole ribonucleotide


5-Amino-4-carboxyamino-imidazole ribonucleotide
Aspartate
8
ATP
ADP + P_i
N-Succinylo-5-aminoimidazole-4-carboxamide ribonucleotide (SAICAR)

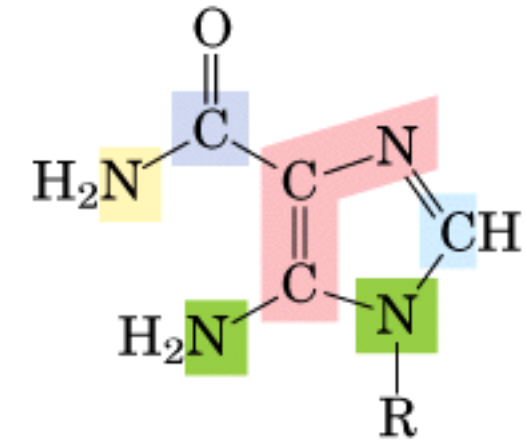


N-Succinylo-5-aminoimidazole-4-carboxamide ribonucleotide (SAICAR)

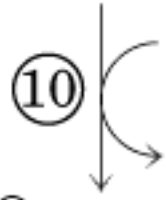
9 → Fumarate

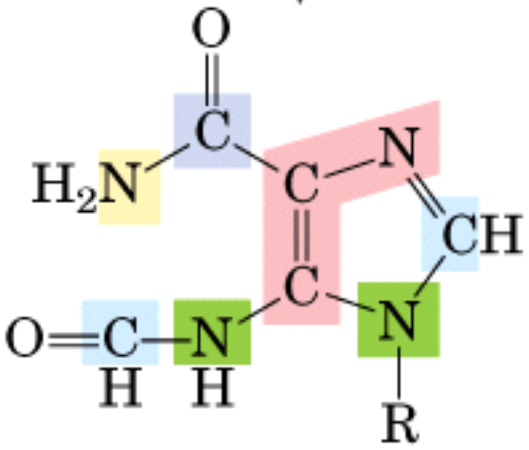


5-Aminoimidazole-4-carboxamide ribonucleotide (AICAR)

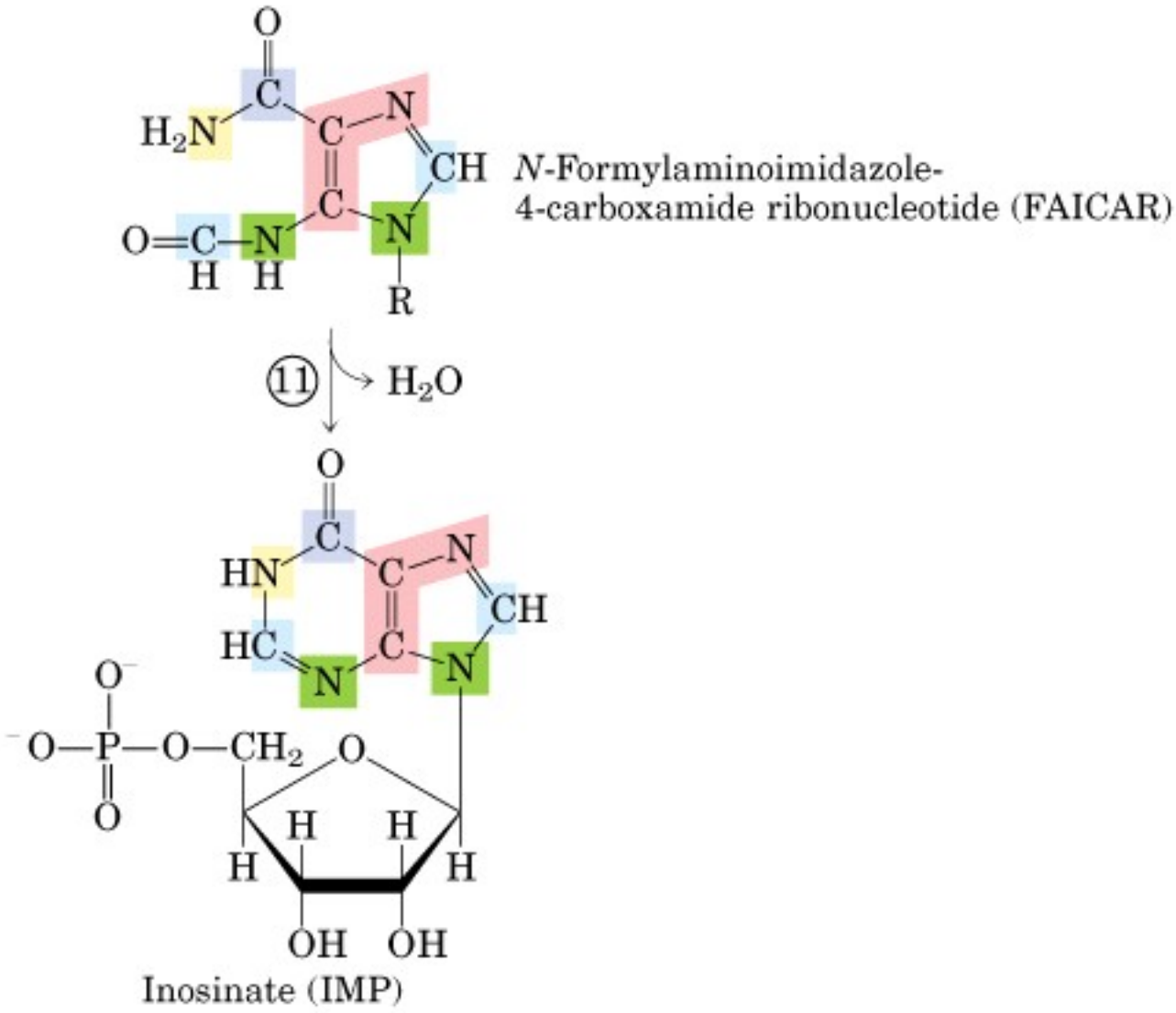


5-Aminoimidazole-4-carboxamide ribonucleotide (AICAR)





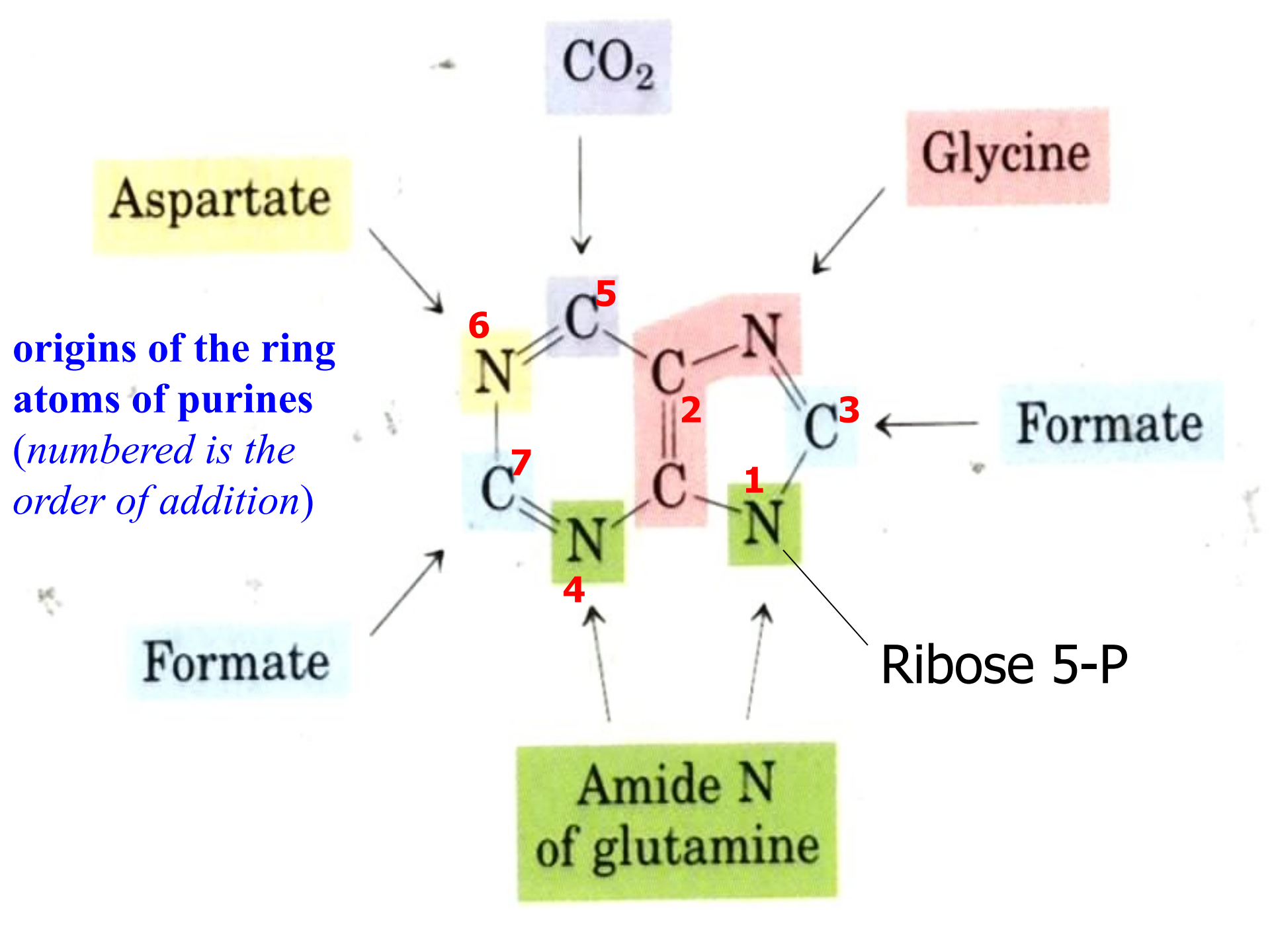
N-Formylaminoimidazole-4-carboxamide ribonucleotide (FAICAR)

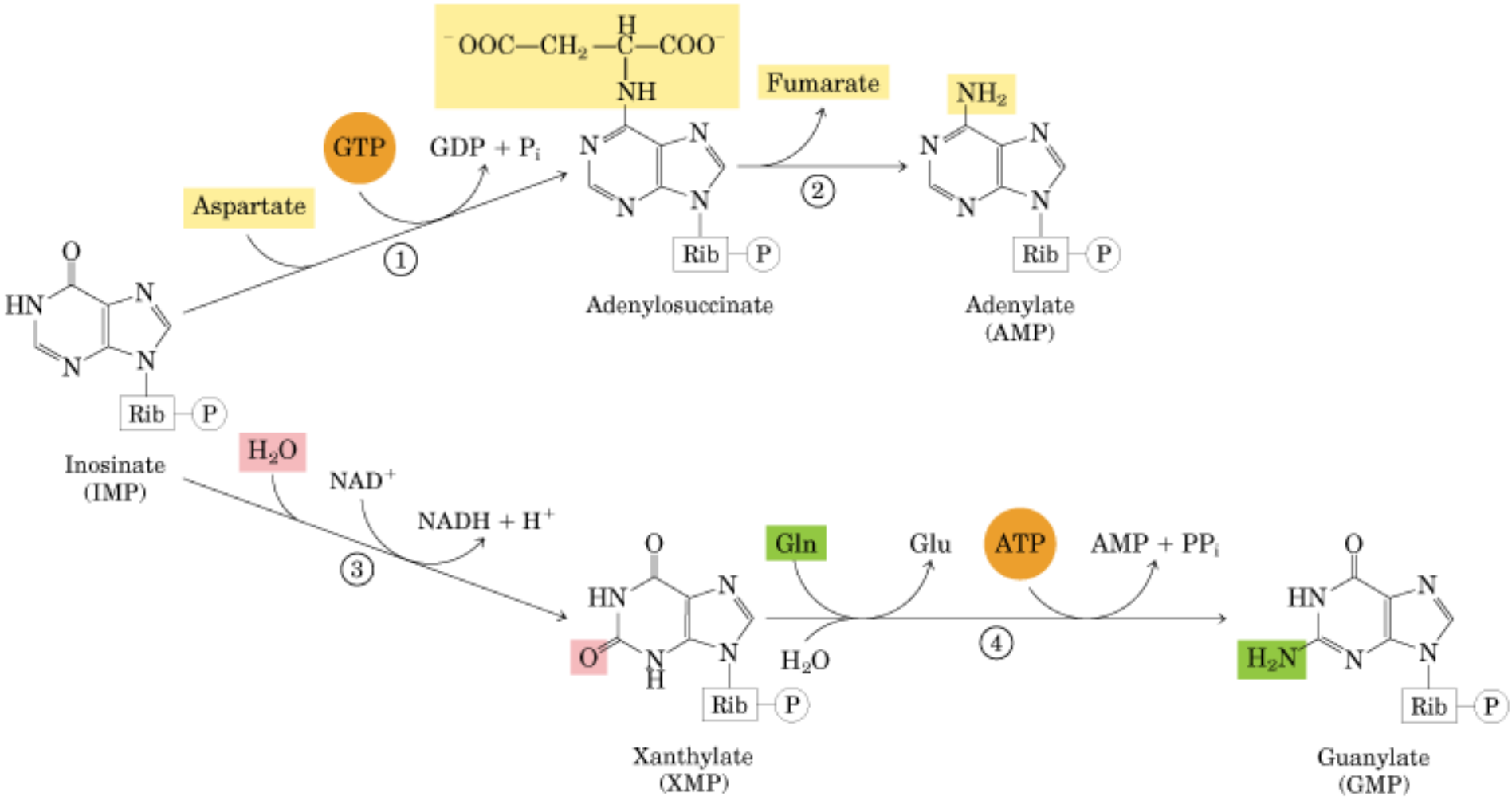


N-Formylaminoimidazole-
4-carboxamide ribonucleotide (FAICAR)

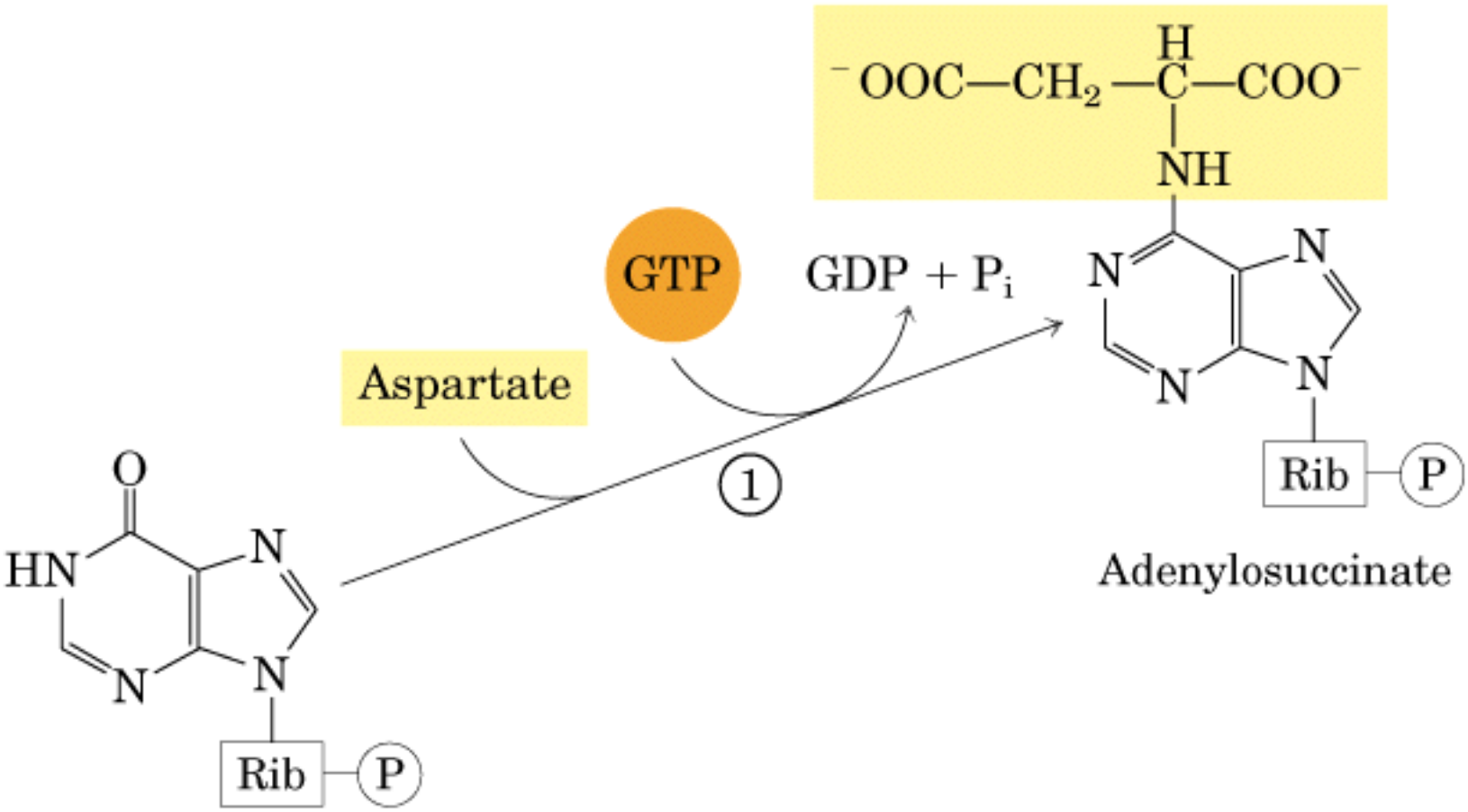
Inosinate (IMP)

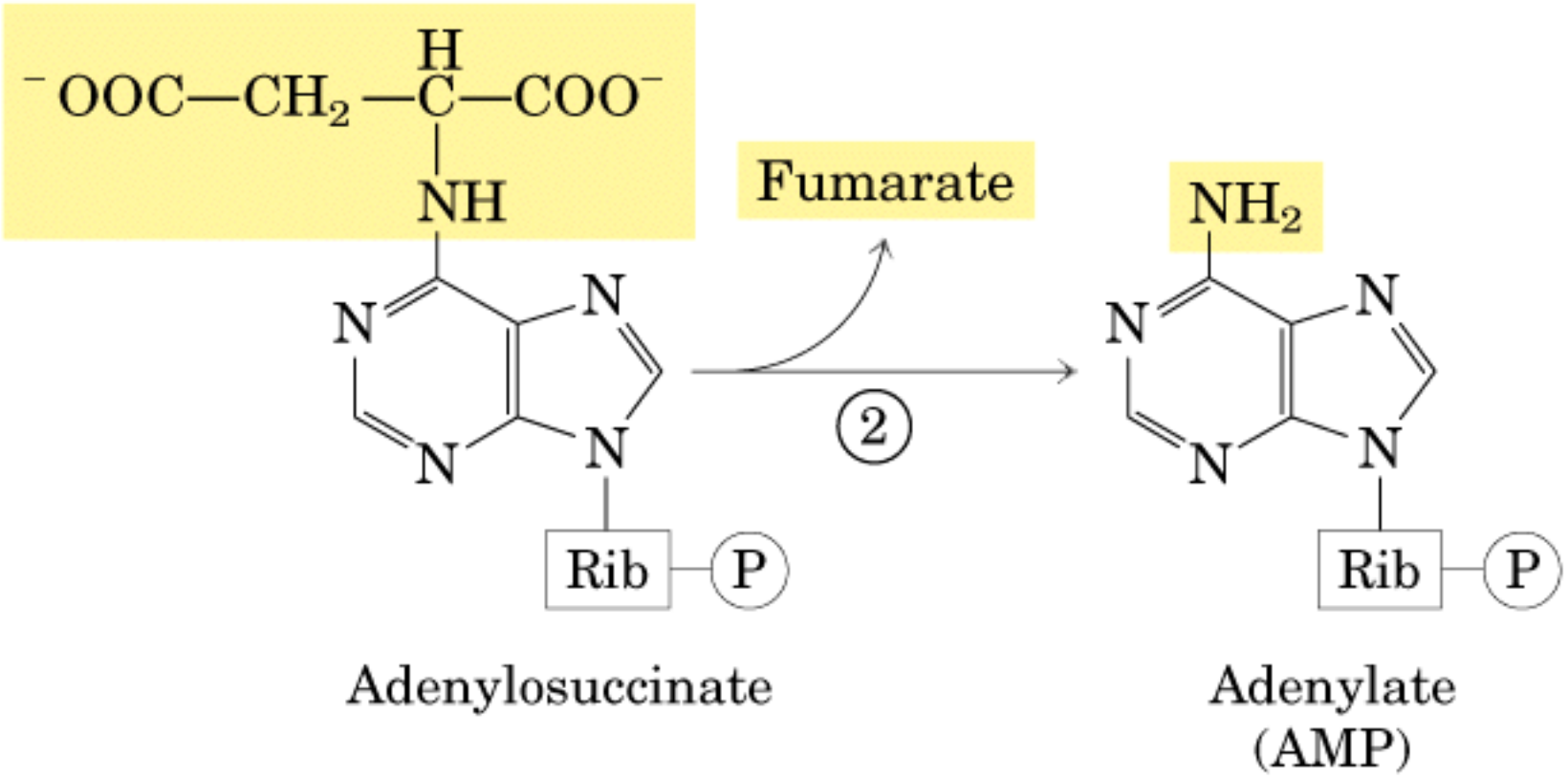
origins of the ring atoms of purines (*numbered is the order of addition*)

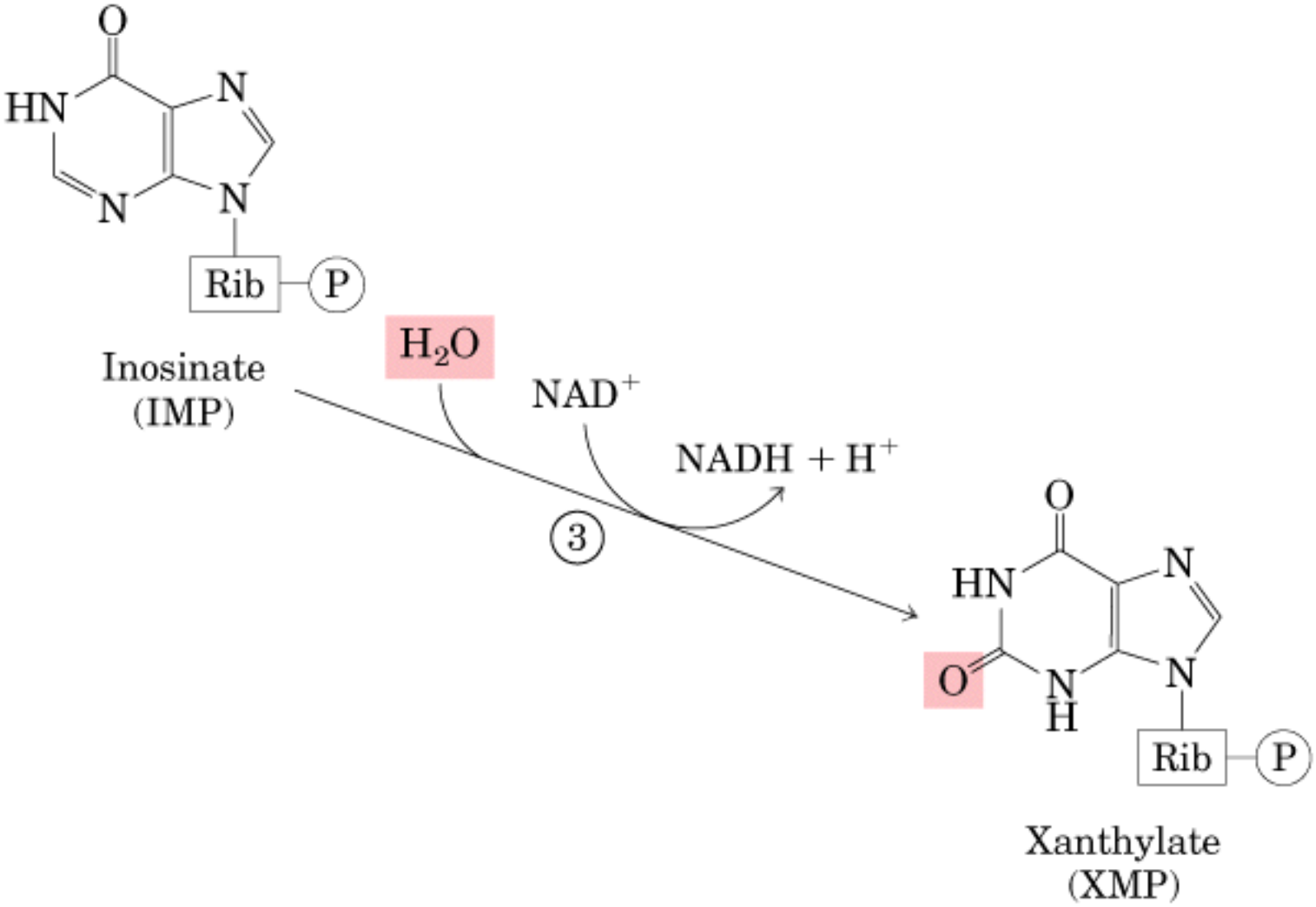


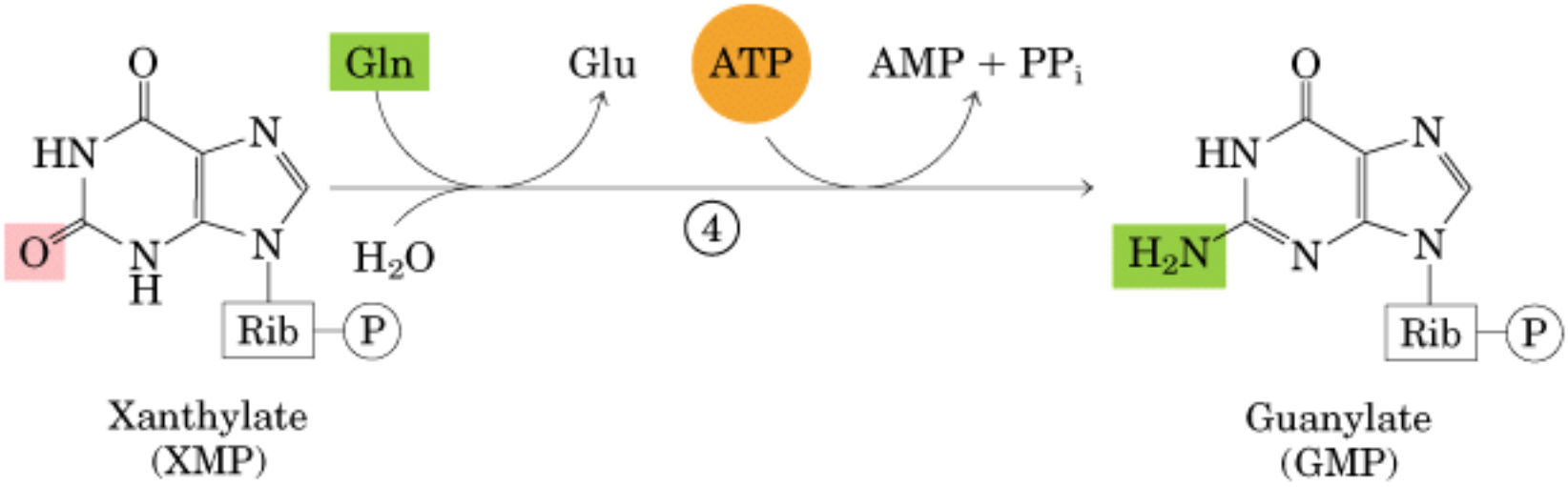

$^{-}OOC—CH_2—\underset{\underset{NH}{|}}{\overset{H}{\overset{|}{C}}}—COO^{-}$
Fumarate
NH_2
GTP
$GDP + P_i$
Aspartate
1
2
Rib
P
Adenylosuccinate
Adenylate
(AMP)
Inosinate
(IMP)
H_2O
NAD^+
$NADH + H^+$
3
Gln
Glu
ATP
$AMP + PP_i$
H_2O
4
H_2N
Xanthylate
(XMP)
Guanylate
(GMP)

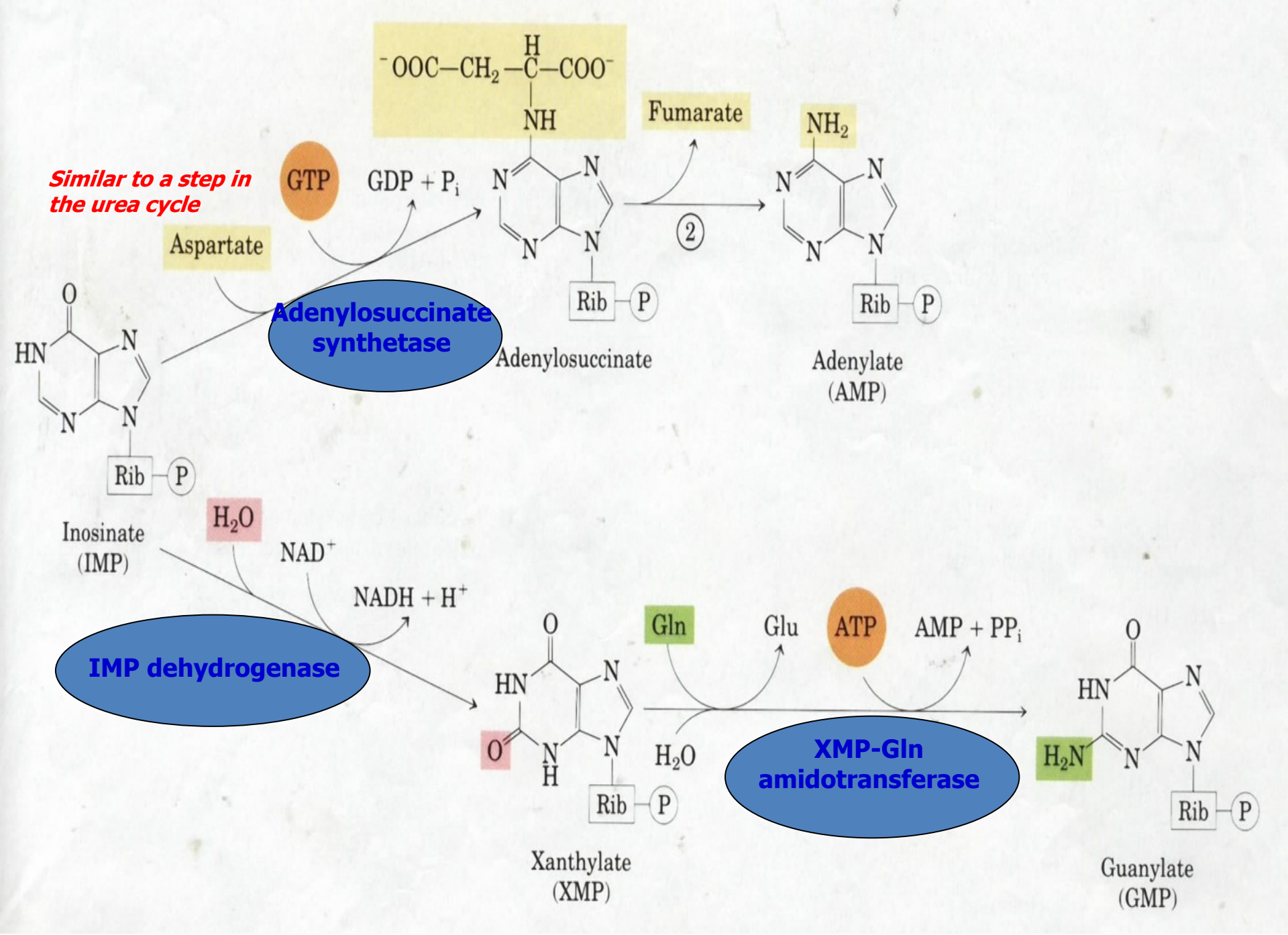
Next


$^-$OOC—CH$_2$—C(H)(NH)—COO$^-$
GTP
GDP + P$_i$
Aspartate
1
Rib
P
Adenylosuccinate
HN
O
N
Rib
P


⁻OOC—CH2—C—COO⁻
H
NH
Fumarate
NH2
N
N
N
N
Rib
P
2
Adenylosuccinate
Adenylate
(AMP)

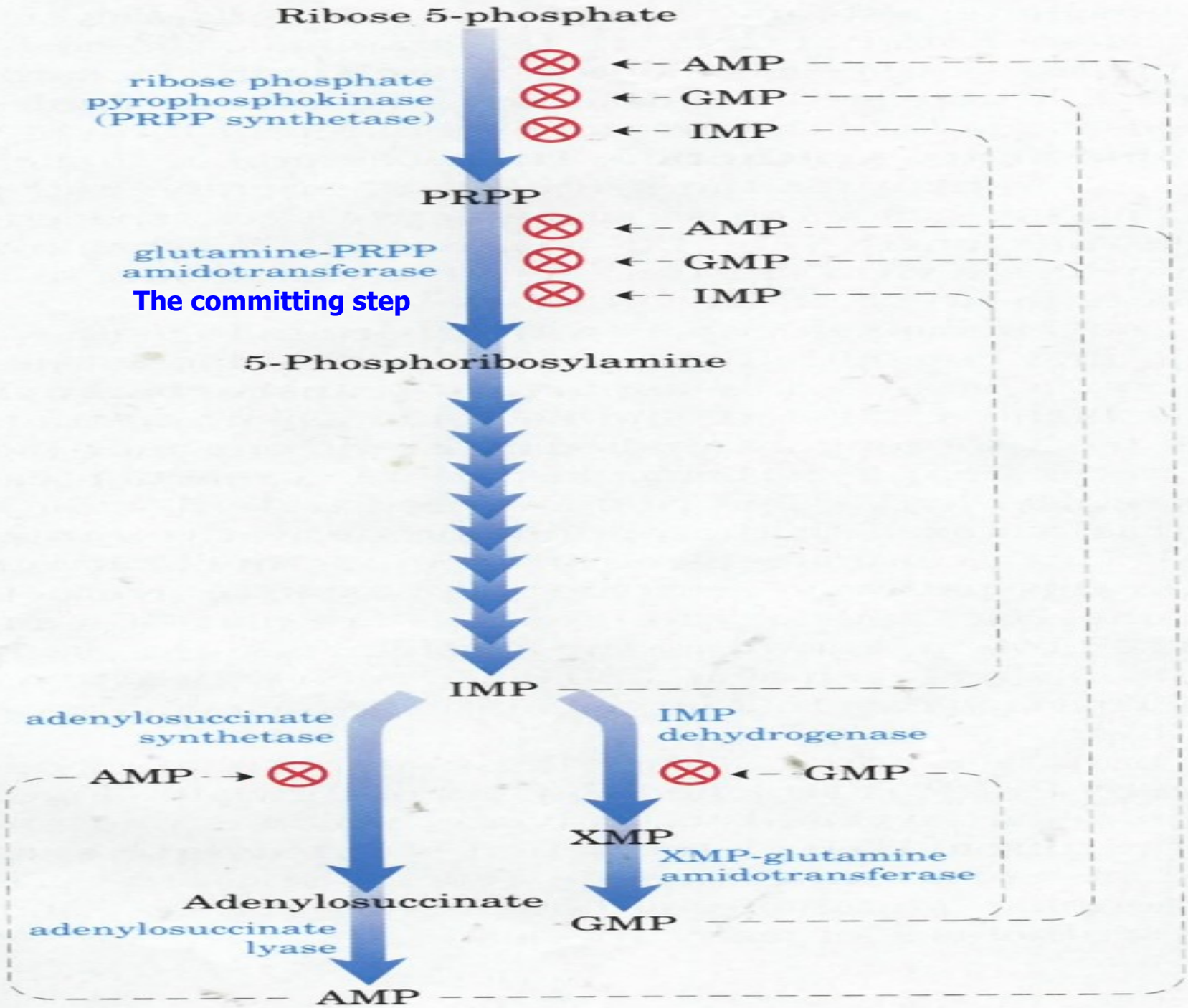

O
HN
N
N
N
Rib
P
Inosinate
(IMP)
H_2O
NAD^+
$NADH + H^+$
3
O
HN
N
O
N
H
N
Rib
P
Xanthylate
(XMP)


Gln
Glu
ATP
AMP + PPi
H2O
4
HN
N
O
N
H
Rib
P
H2N
Xanthylate
(XMP)
Guanylate
(GMP)


Similar to a step in the urea cycle
GTP
GDP + Pi
Aspartate
Adenylosuccinate synthetase
Fumarate
2
Adenylosuccinate
Adenylate (AMP)
Inosinate (IMP)
H2O
NAD+
NADH + H+
IMP dehydrogenase
Gln
Glu
ATP
AMP + PPi
H2O
XMP-Gln amidotransferase
Xanthylate (XMP)
Guanylate (GMP)

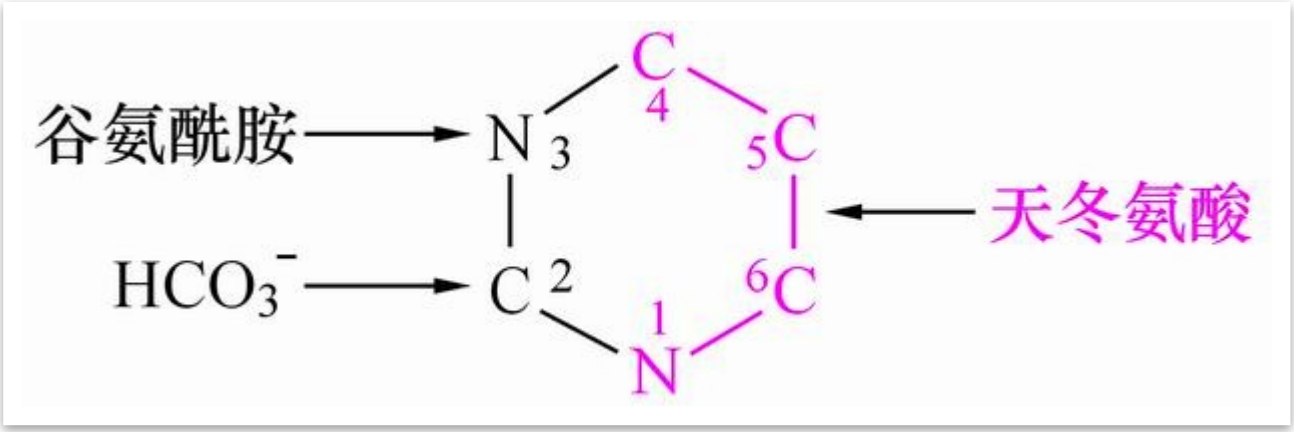
The biosynthesis of AMP and GMP is regulated mainly by sequential feedback inhibition (*no covalent regulation*)

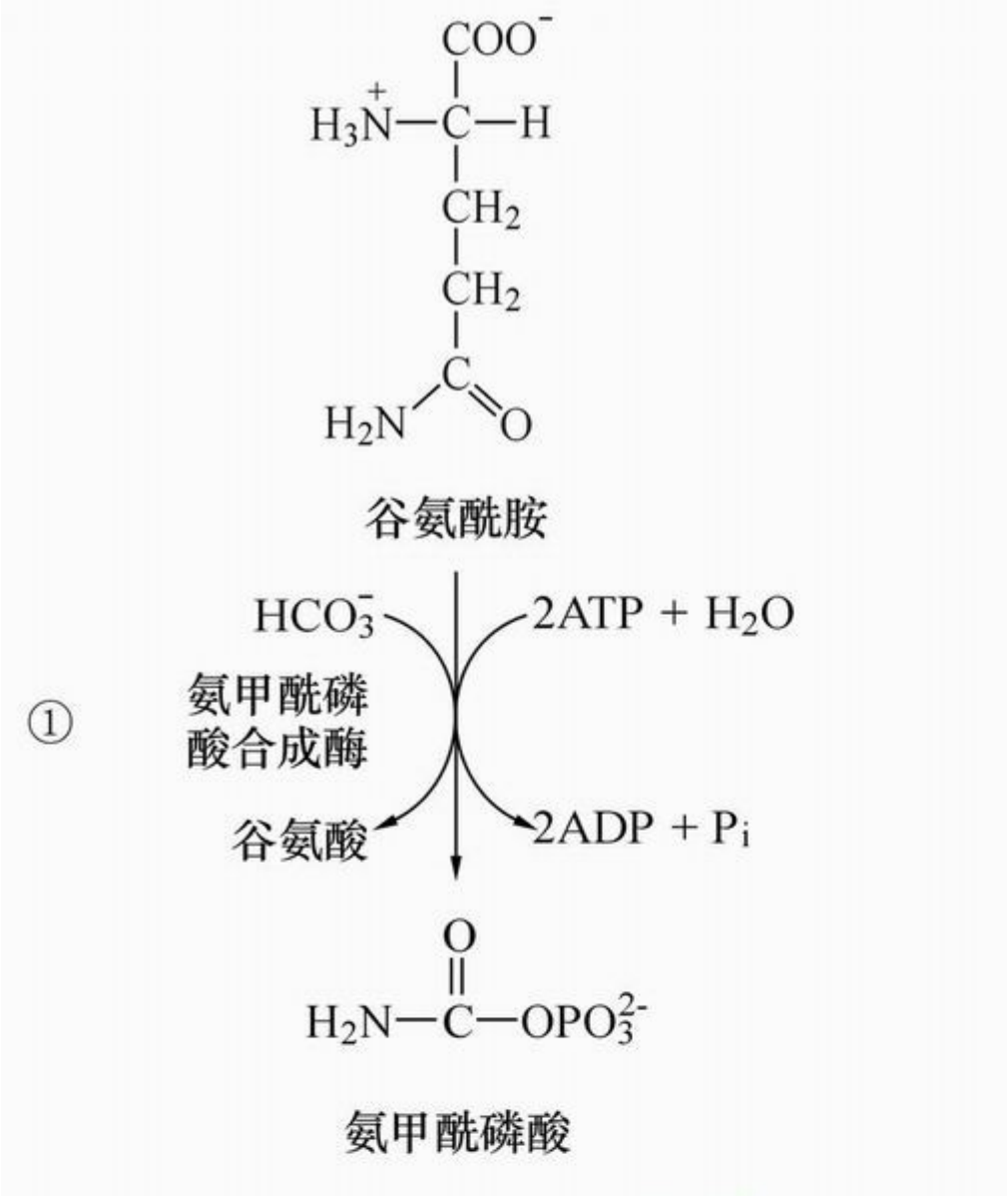


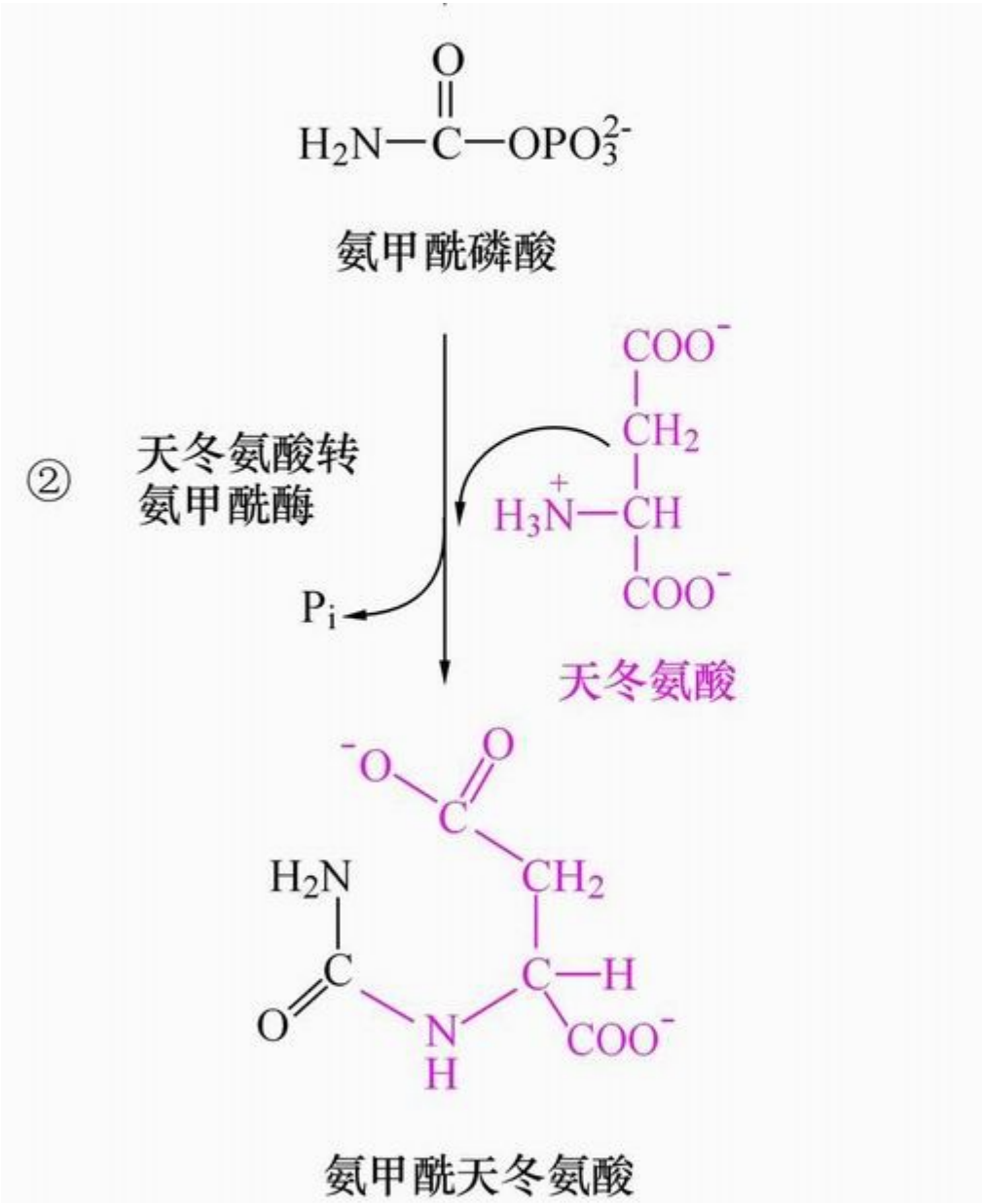

Ribose 5-phosphate
ribose phosphate pyrophosphokinase (PRPP synthetase)
AMP
GMP
IMP
PRPP
glutamine-PRPP amidotransferase
The committing step
AMP
GMP
IMP
5-Phosphoribosylamine
IMP
adenylosuccinate synthetase
AMP
IMP dehydrogenase
GMP
XMP
XMP-glutamine amidotransferase
Adenylosuccinate
adenylosuccinate lyase
GMP
AMP

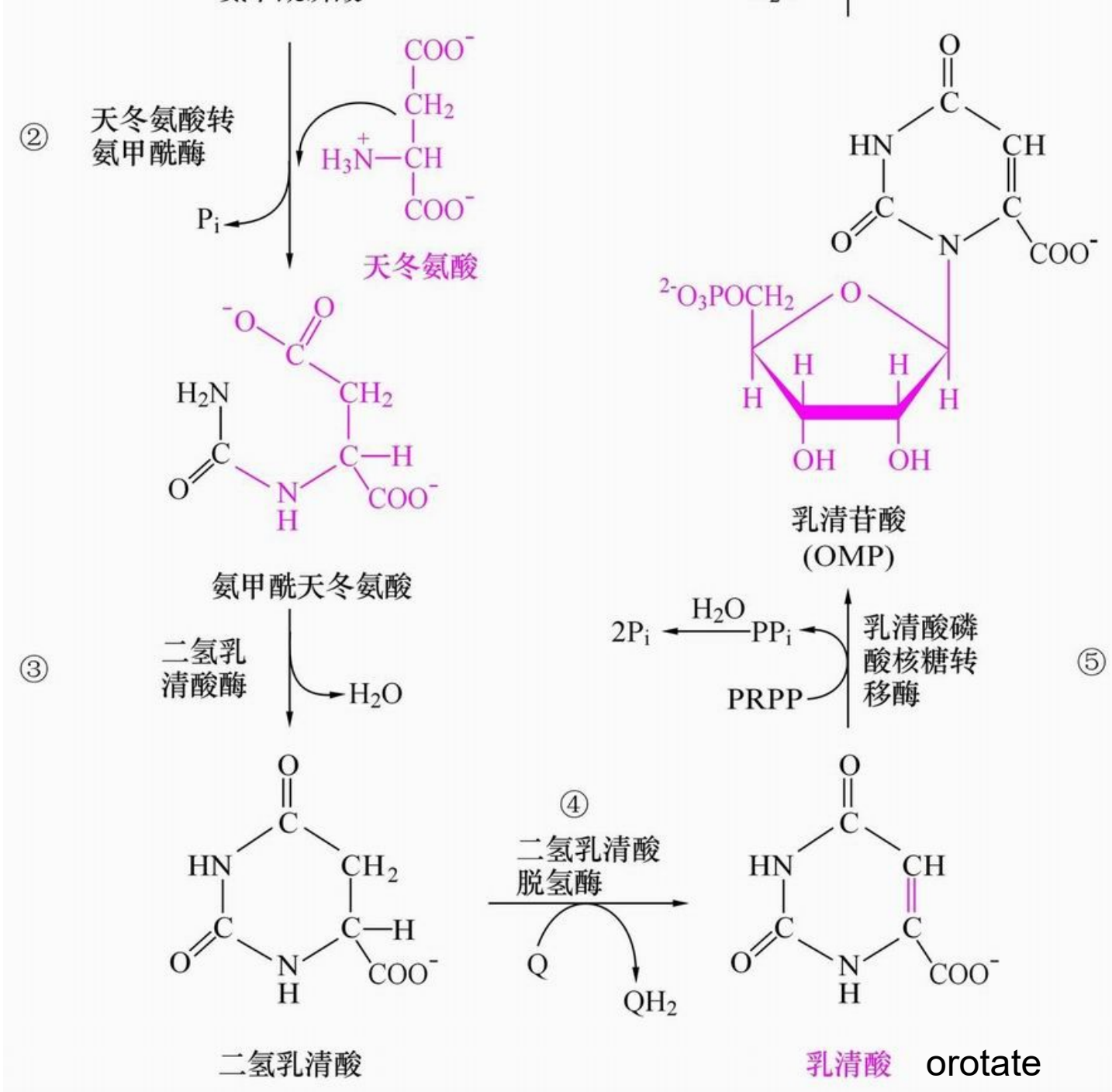
The *de novo* biosynthesis of pyrimidine nucleotides: the base ring (as orotate) is first assembled before being attached to the phosphoribosyl group to form orotidylate

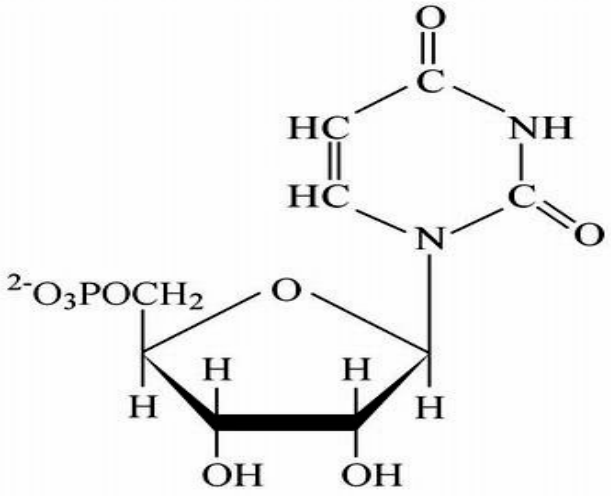



谷氨酰胺
HCO_3^-
N 3
C 2
C 4
5C
6C
1 N
天冬氨酸

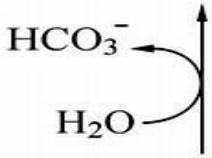

谷氨酰胺
①
HCO_3^-
2ATP + H_2O
氨甲酰磷
酸合成酶
谷氨酸
2ADP + P_i
氨甲酰磷酸


氨甲酰磷酸
②
天冬氨酸转
氨甲酰酶
Pi
天冬氨酸
氨甲酰天冬氨酸

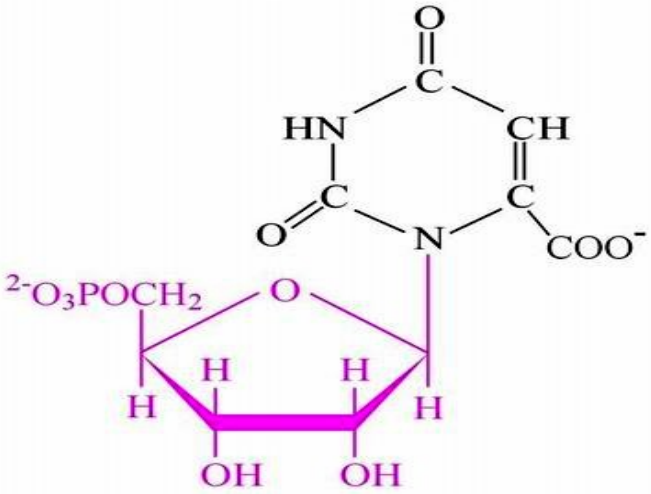

②
天冬氨酸转
氨甲酰酶
天冬氨酸
氨甲酰天冬氨酸
③
二氢乳
清酸酶
二氢乳清酸
④
二氢乳清酸
脱氢酶
乳清酸
⑤
乳清酸磷
酸核糖转
移酶
乳清苷酸
(OMP)
orotate



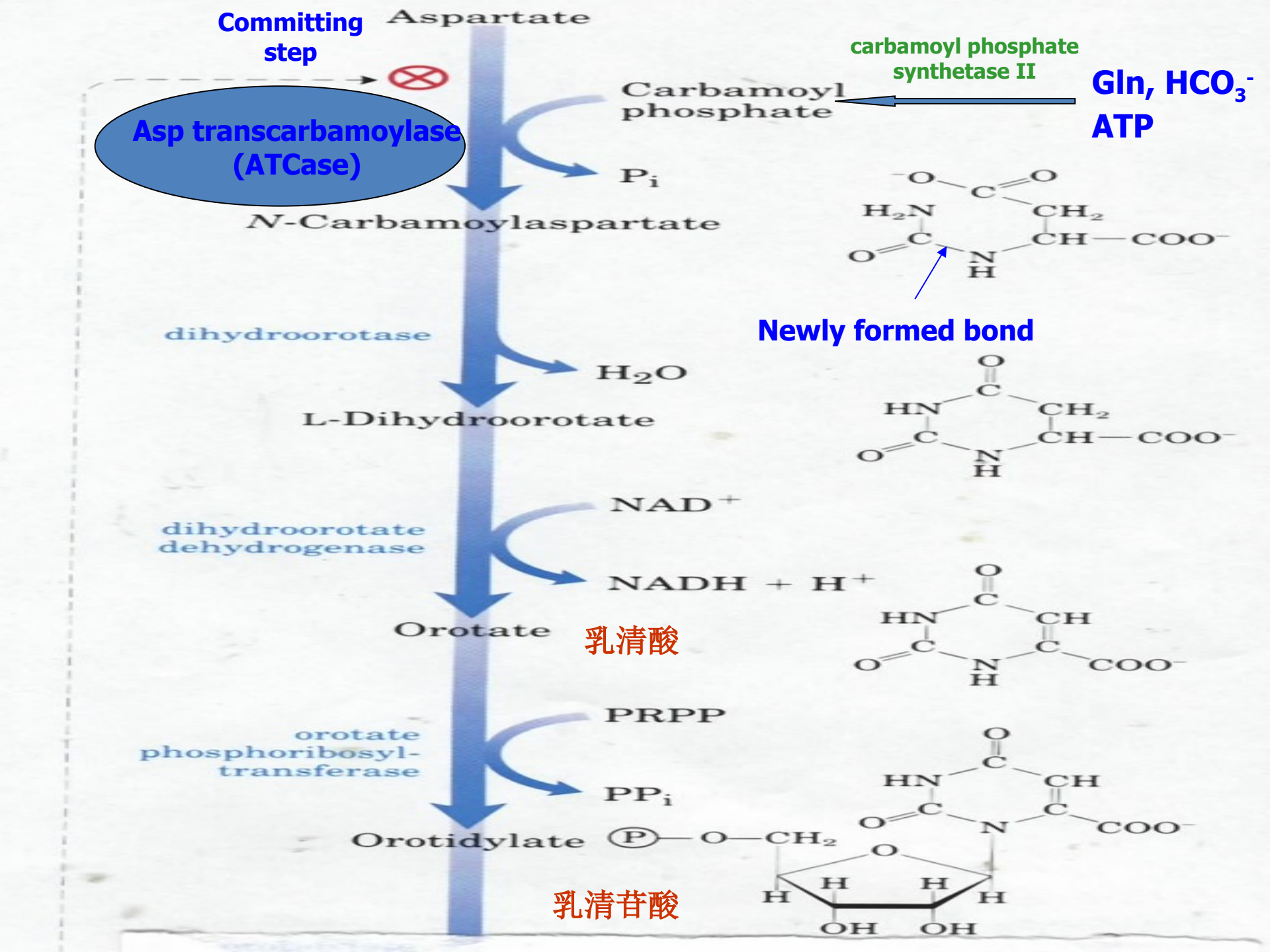
尿苷酸(UMP)



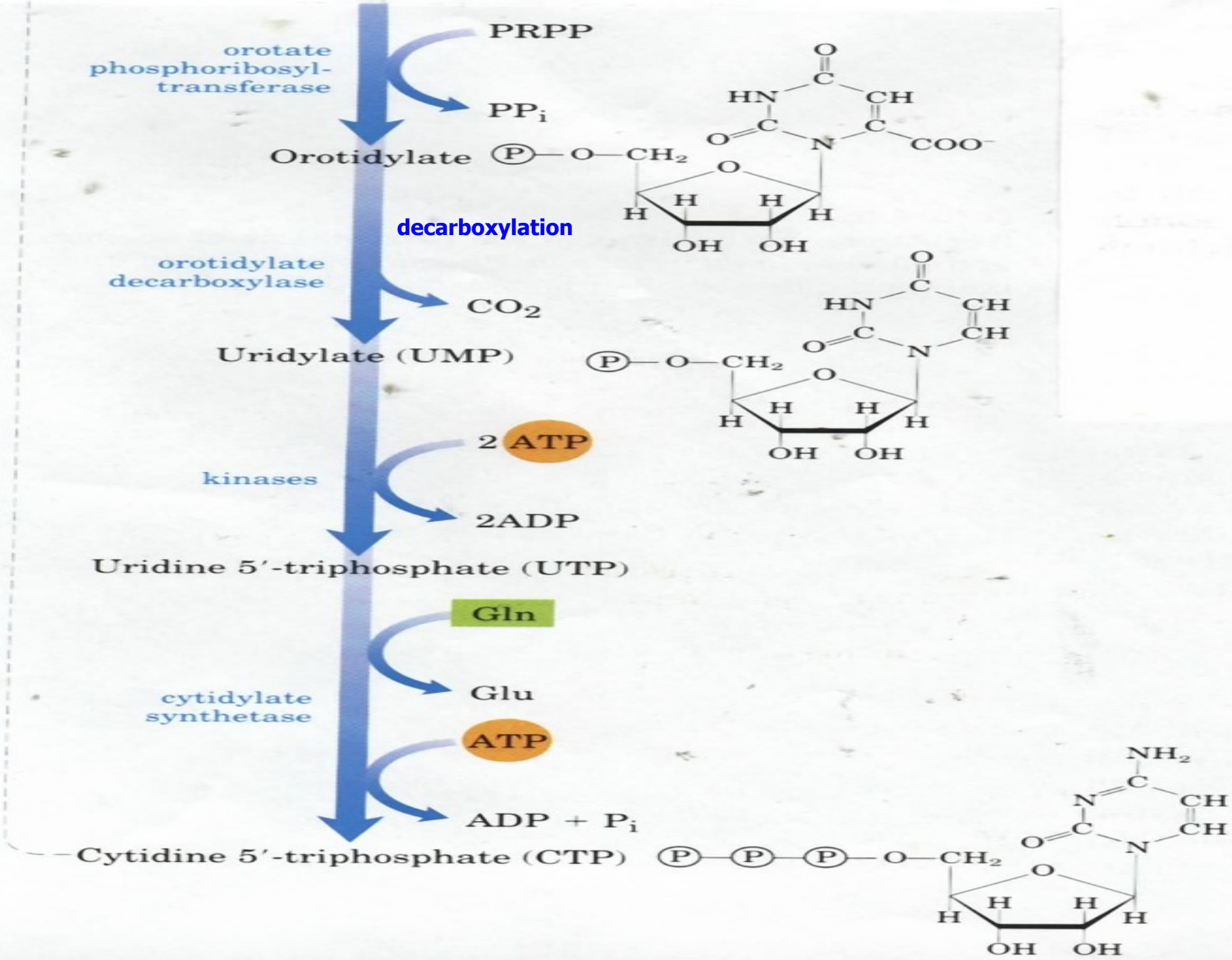
OMP脱羧酶 ⑥



乳清苷酸
(OMP)

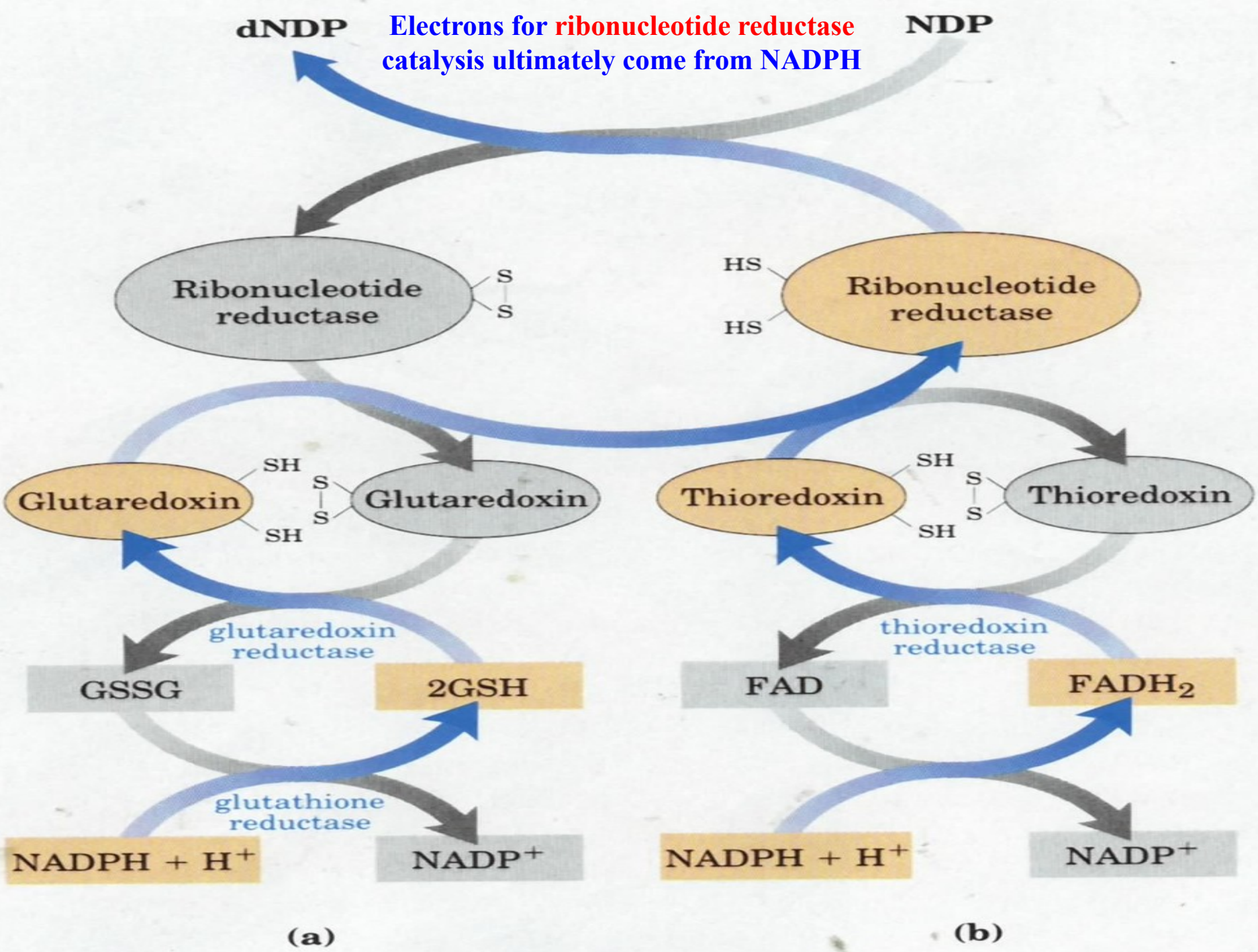

Committing step
Aspartate
carbamoyl phosphate synthetase II
Gln, HCO_3^-
ATP
Asp transcarbamoylase (ATCase)
Carbamoyl phosphate
P_i
N-Carbamoylaspartate
Newly formed bond
dihydroorotase
H_2O
L-Dihydroorotate
NAD^+
dihydroorotate dehydrogenase
$NADH + H^+$
Orotate
乳清酸
PRPP
orotate phosphoribosyl-transferase
PP_i
Orotidylate
乳清苷酸

UTP and CTP are then derived from orotidylate *via* decarboxylation and amination reactions


PRPP
orotate phosphoribosyl-transferase
PP_i
Orotidylate
decarboxylation
orotidylate decarboxylase
CO_2
Uridylate (UMP)
2 ATP
kinases
2ADP
Uridine 5′-triphosphate (UTP)
Gln
Glu
cytidylate synthetase
ATP
ADP + P_i
Cytidine 5′-triphosphate (CTP)

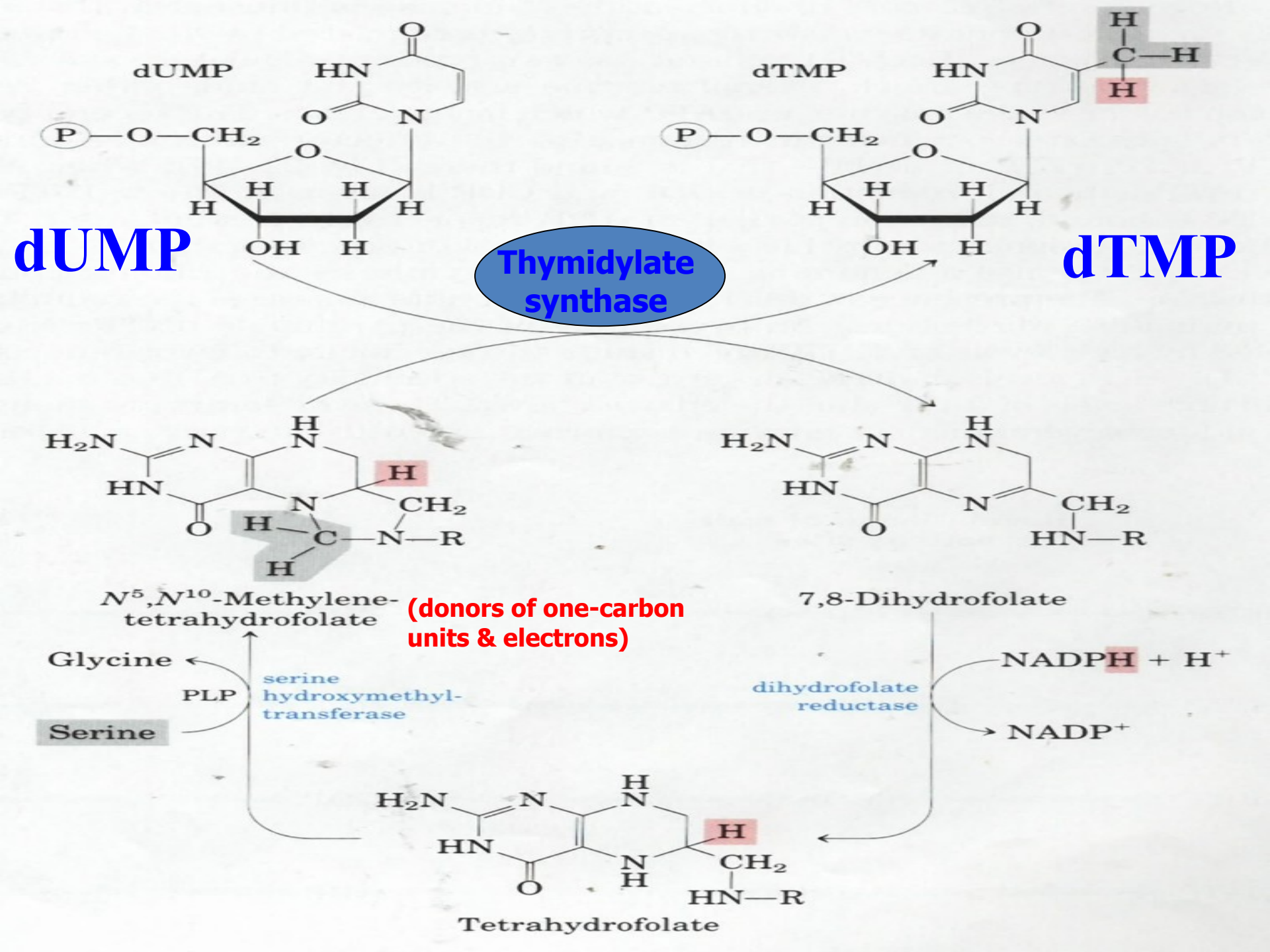
All dNDPs are derived from NDPs via the catalysis of a common ribonucleotide reductase.

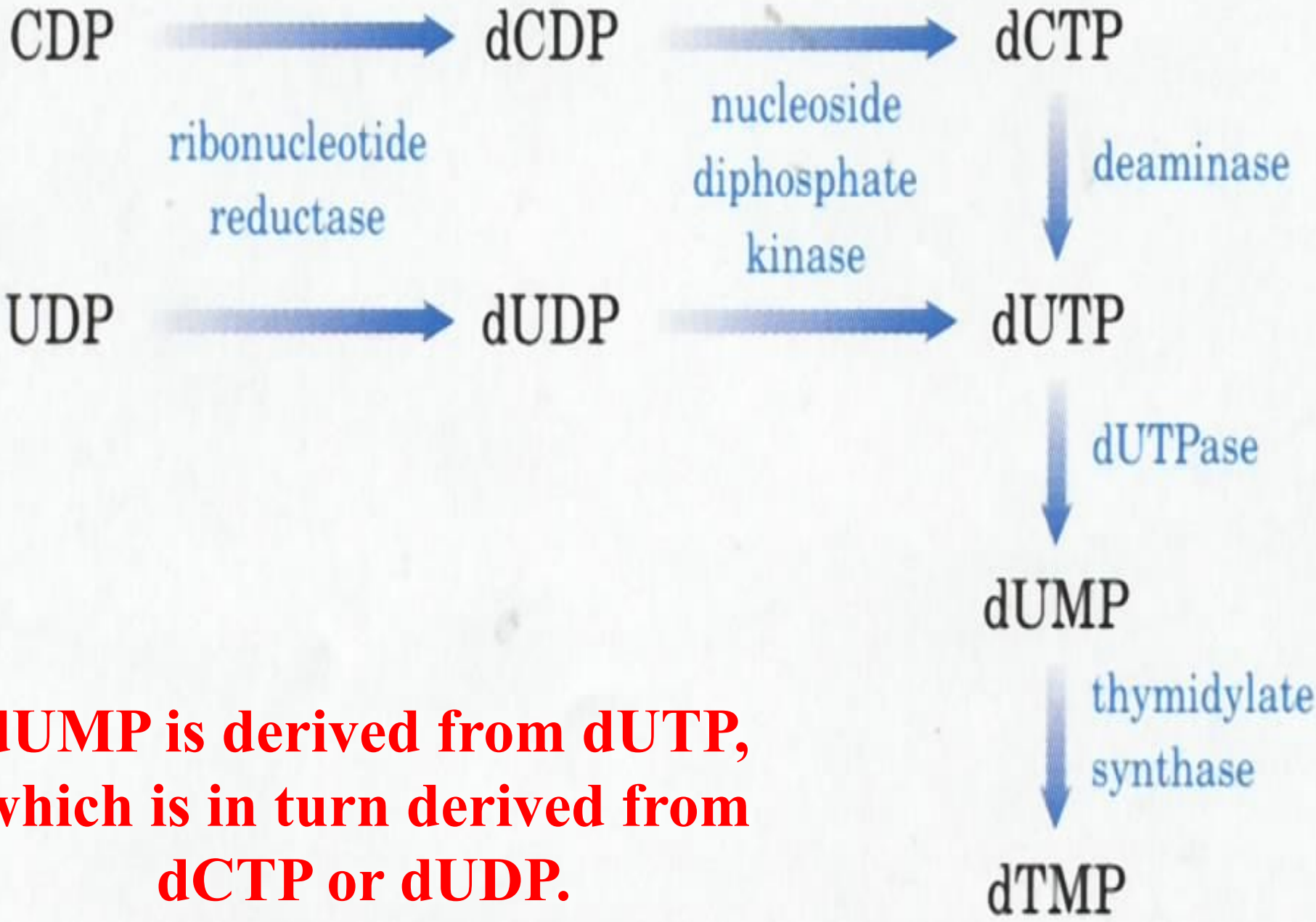



dNDP
Electrons for ribonucleotide reductase catalysis ultimately come from NADPH
NDP
Ribonucleotide reductase
S
S
HS
HS
Ribonucleotide reductase
Glutaredoxin
SH
SH
S
S
Glutaredoxin
Thioredoxin
SH
SH
S
S
Thioredoxin
glutaredoxin reductase
thioredoxin reductase
GSSG
2GSH
FAD
FADH2
glutathione reductase
NADPH + H+
NADP+
NADPH + H+
NADP+
(a)
(b)

dTMP is synthesized by the methylation of dUMP

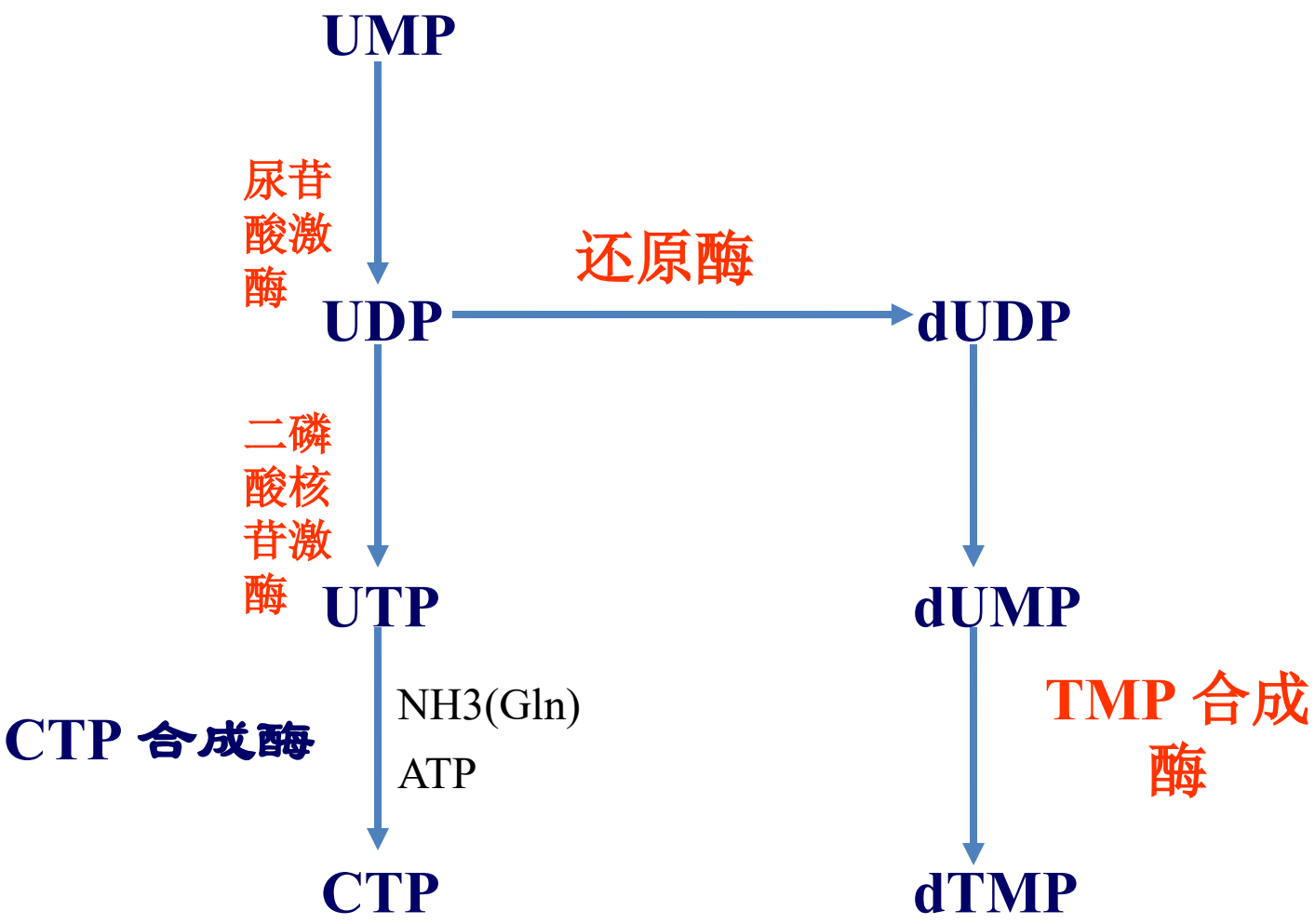



dUMP
dTMP
dUMP
Thymidylate synthase
dTMP
N^5,N^{10}-Methylene-tetrahydrofolate
(donors of one-carbon units & electrons)
7,8-Dihydrofolate
Glycine
PLP
serine hydroxymethyl-transferase
Serine
dihydrofolate reductase
NADPH + H+
NADP+
Tetrahydrofolate



dUMP is derived from dUTP, which is in turn derived from dCTP or dUDP.

UMP 转变为 CTP 和 dTMP



Biosynthesis Nucleotide

IMP (Inosinate 肌苷酸、次黄嘌呤核苷酸) ------**AMP**, **GMP**

O (orotate 乳清酸） ------OMP(乳清苷酸)-------**UMP**----UTP------**CTP**

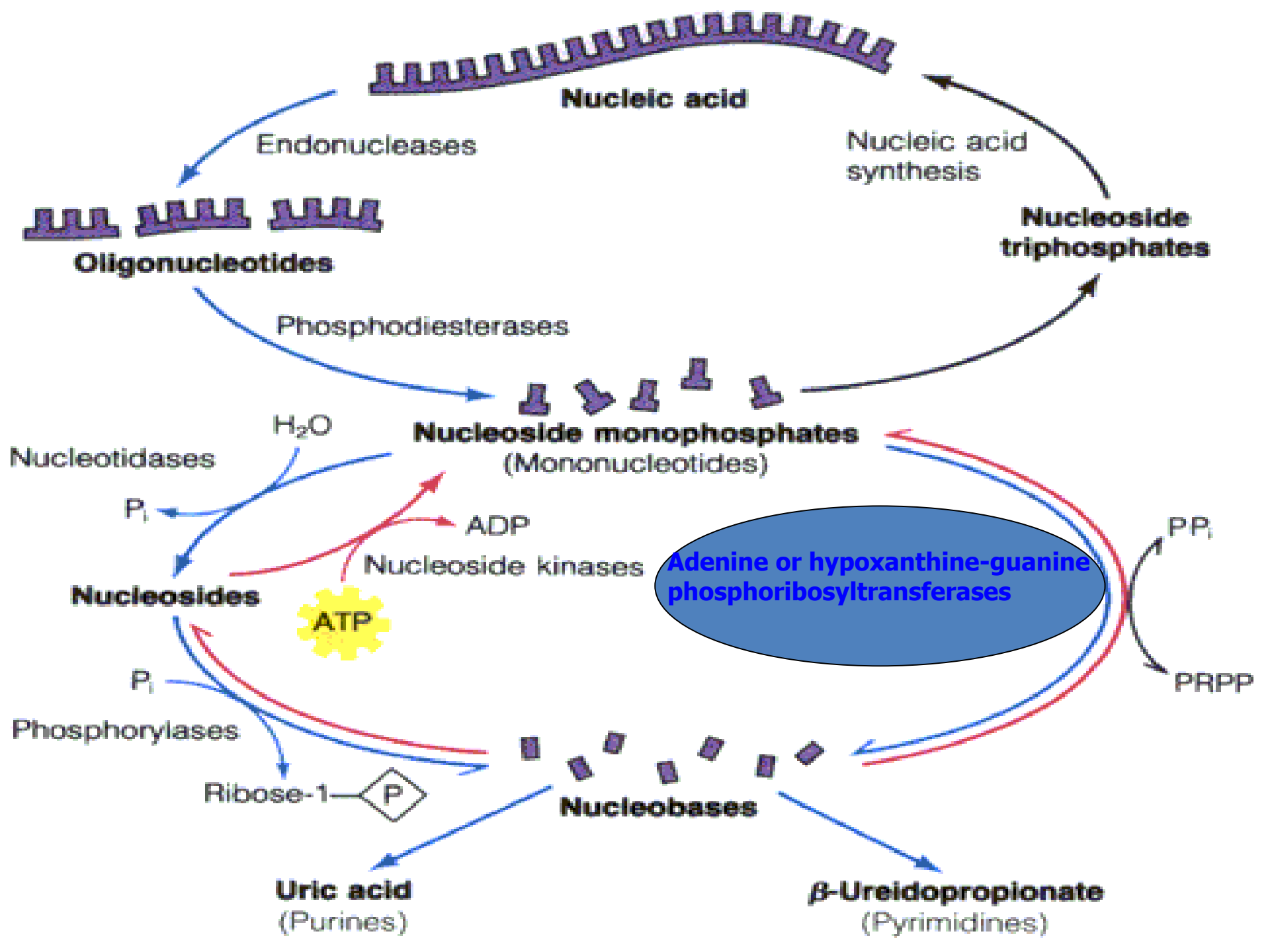
Biosynthesis Deoxynucleotide

ADP-----**dADP**
GDP----**dGDP**
CDP-----**dCDP**

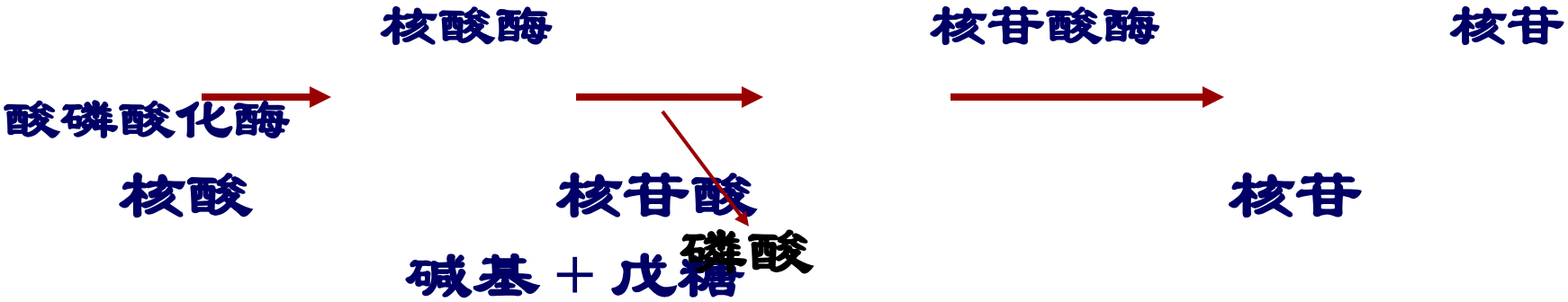
UDP----dUDP----dUMP----**dTMP**

Purine and pyrimidine bases can be reconverted to nucleotides via the salvage pathway




Nucleic acid
Endonucleases
Nucleic acid synthesis
Oligonucleotides
Nucleoside triphosphates
Phosphodiesterases
Nucleoside monophosphates
(Mononucleotides)
H2O
Nucleotidases
Pi
ADP
Nucleoside kinases
Nucleosides
ATP
Adenine or hypoxanthine-guanine phosphoribosyltransferases
PPi
PRPP
Pi
Phosphorylases
Ribose-1
P
Nucleobases
Uric acid
(Purines)
β-Ureidopropionate
(Pyrimidines)

核酸的酶促降解



Next

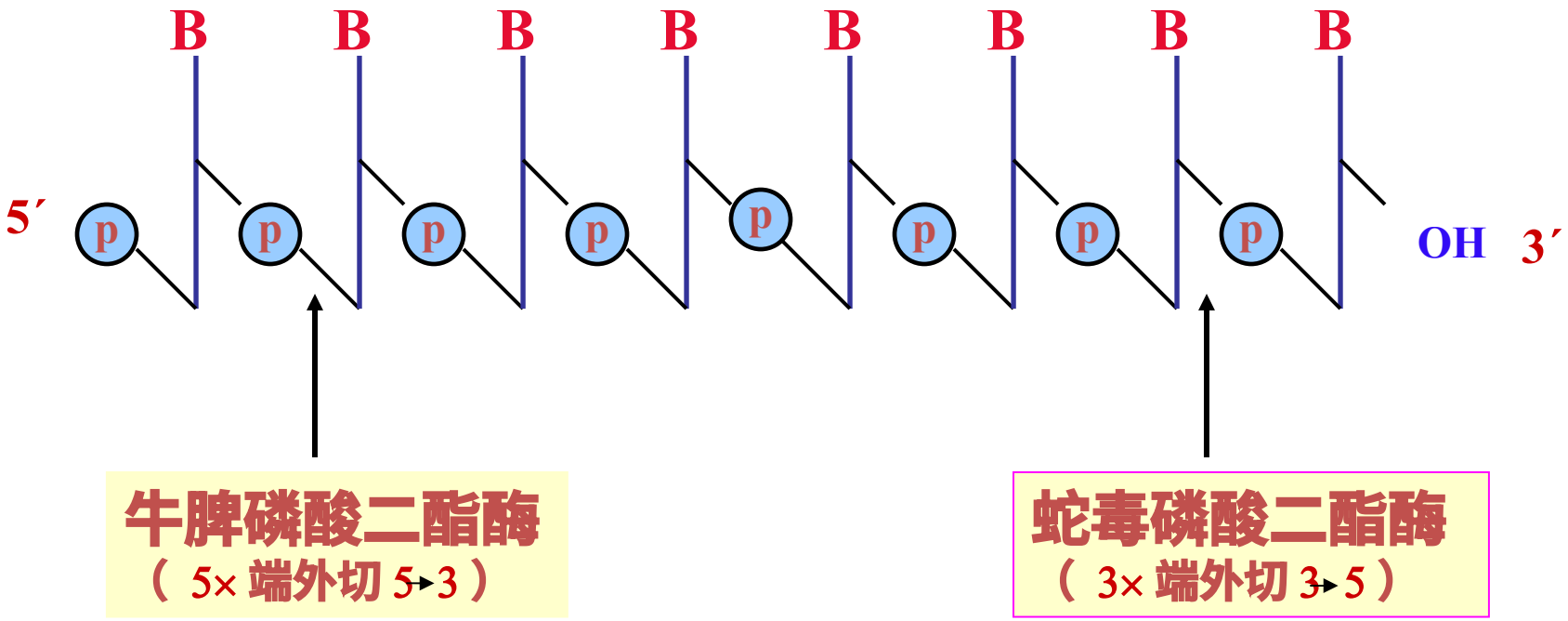
核　　酸　　酶

1、核酸酶的分类

（1）根据对底物的专一性分为
- 核糖核酸酶（RNase）
- 脱氧核糖核酸酶（DNase）
- 非特异性核酸酶

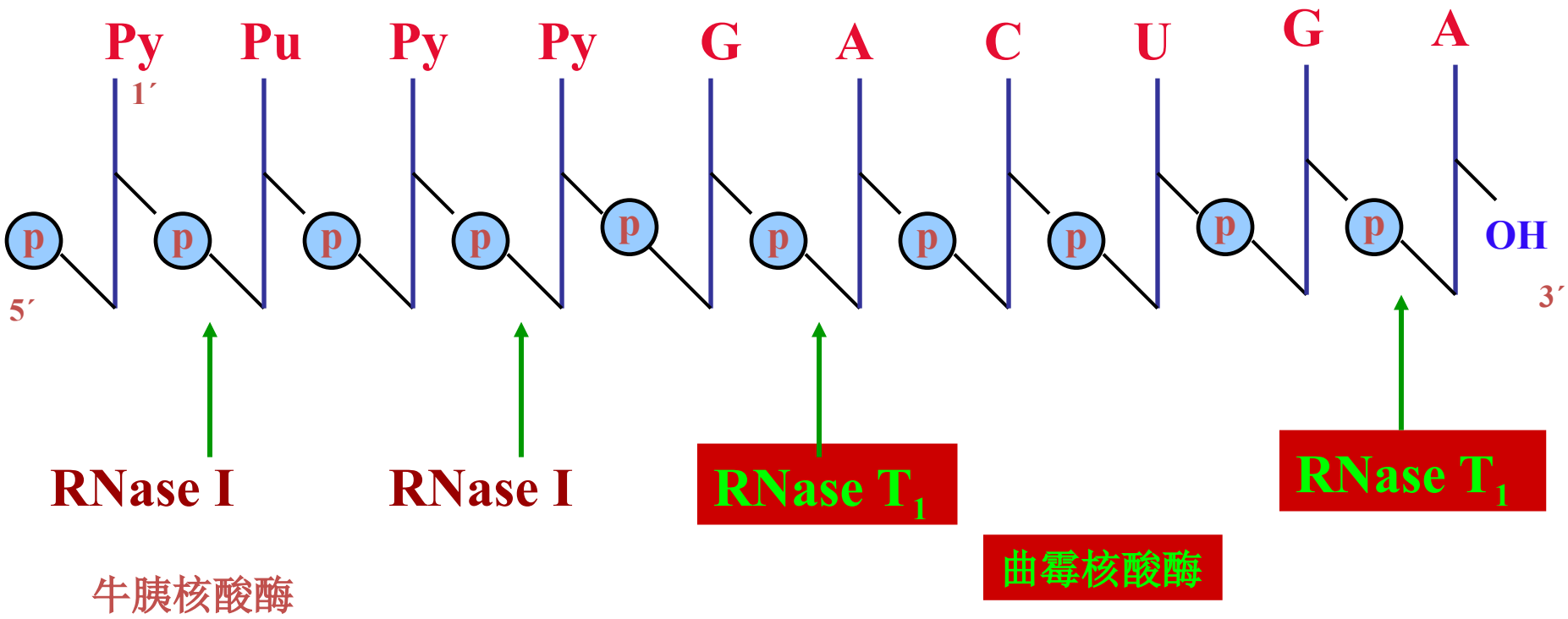
（2）根据切割位点分为
- 核酸内切酶
- 核酸外切酶
- 核酸内切兼外切酶

1、外切核酸酶对核酸的水解位点



非特异性核酸酶，作用于 DNA 、 RNA

2、内切核酸酶对 RNA 的水解位点示意图

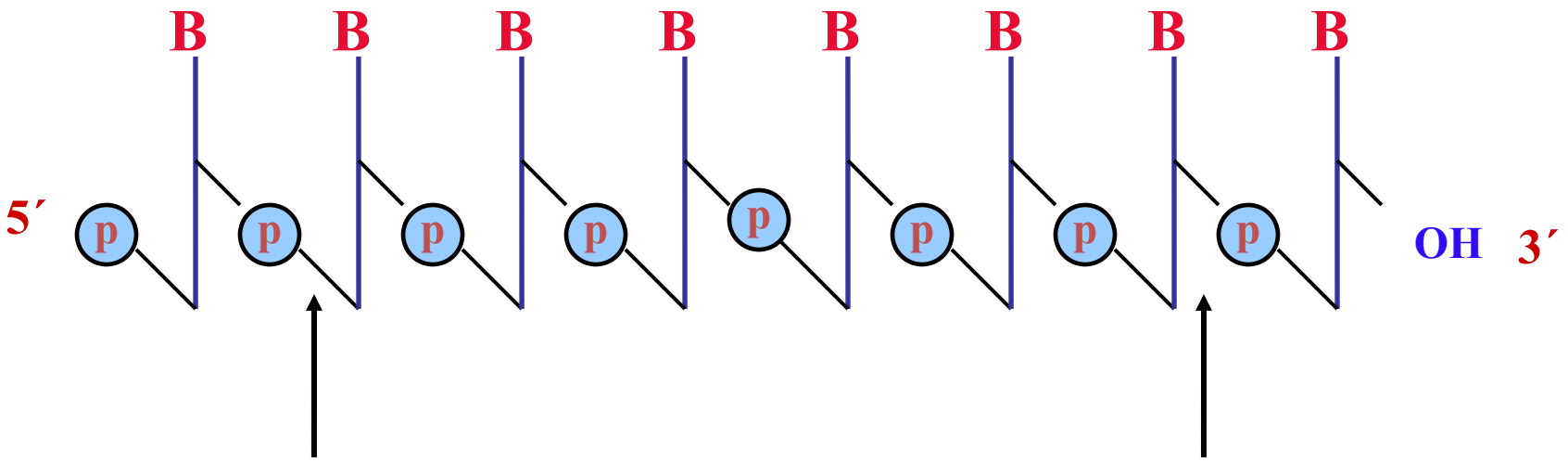


Pu ： 嘌呤　　Py ： 嘧啶

特异性核酸酶，只作用于 RNA 或 DNA

3、脱氧核糖核酸酶

特异性核酸酶，只作用于 DNA



牛脾脱氧核糖核酸酶
（ DNase II ）

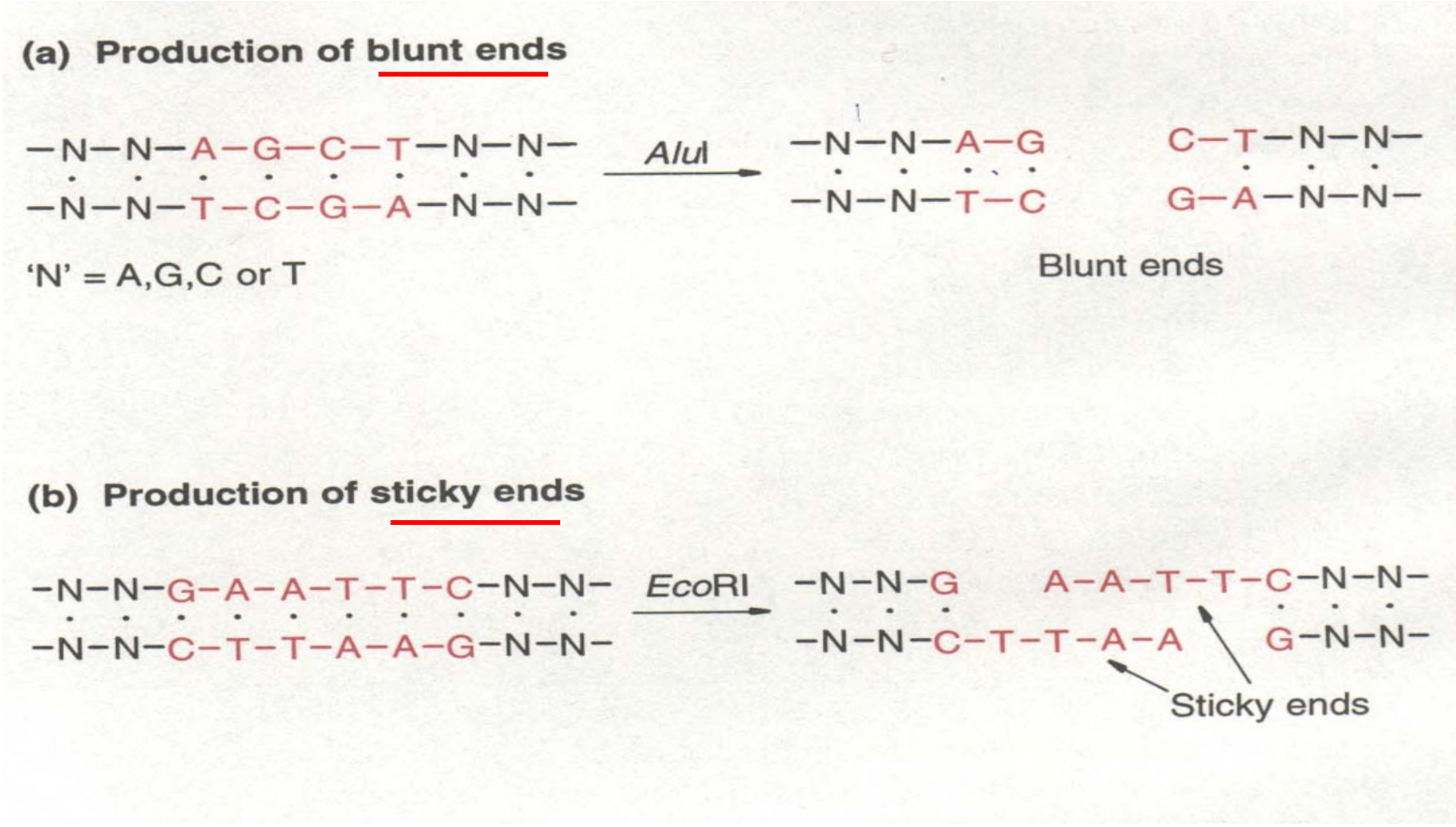
牛胰脱氧核糖核酸酶
（ DNase I ）

限制性内切酶

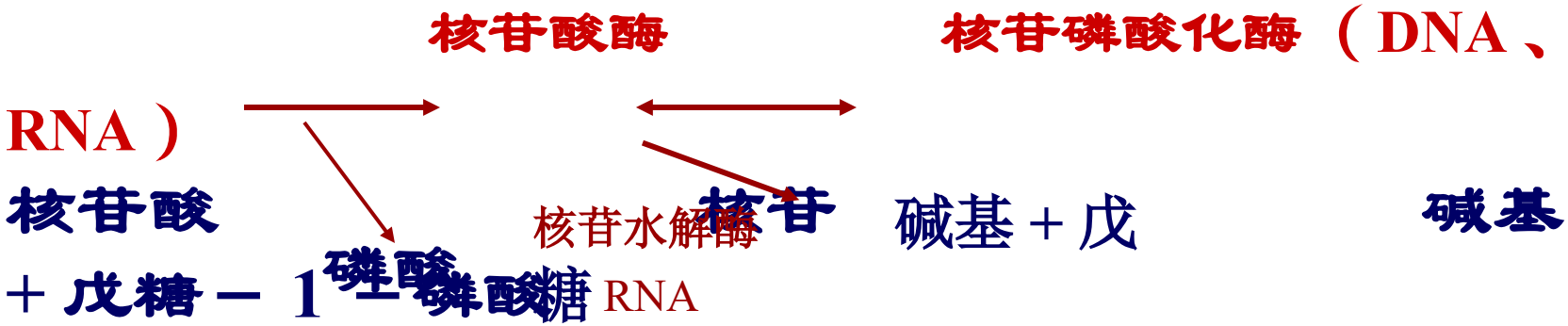
原核生物中存在着一类能识别外源 DNA 双螺旋中 4-8 个碱基对所组成的特异的序列，并在此序列的某位点水解 DNA 双螺旋链，产生粘性末端或平齐末端，这类酶称为限制性内切酶（ restriction endonuclease ）。

具有很强的专一性， 有特异的识别位点，通常具有二重旋转对称性的回文序列 (palindromic sequence)




(a) Production of blunt ends
–N–N–A–G–C–T–N–N–
–N–N–T–C–G–A–N–N–
AluI
–N–N–A–G
–N–N–T–C
C–T–N–N–
G–A–N–N–
'N' = A,G,C or T
Blunt ends
(b) Production of sticky ends
–N–N–G–A–A–T–T–C–N–N–
–N–N–C–T–T–A–A–G–N–N–
EcoRI
–N–N–G
–N–N–C–T–T–A–A
A–A–T–T–C–N–N–
G–N–N–
Sticky ends

核苷酸分解代谢



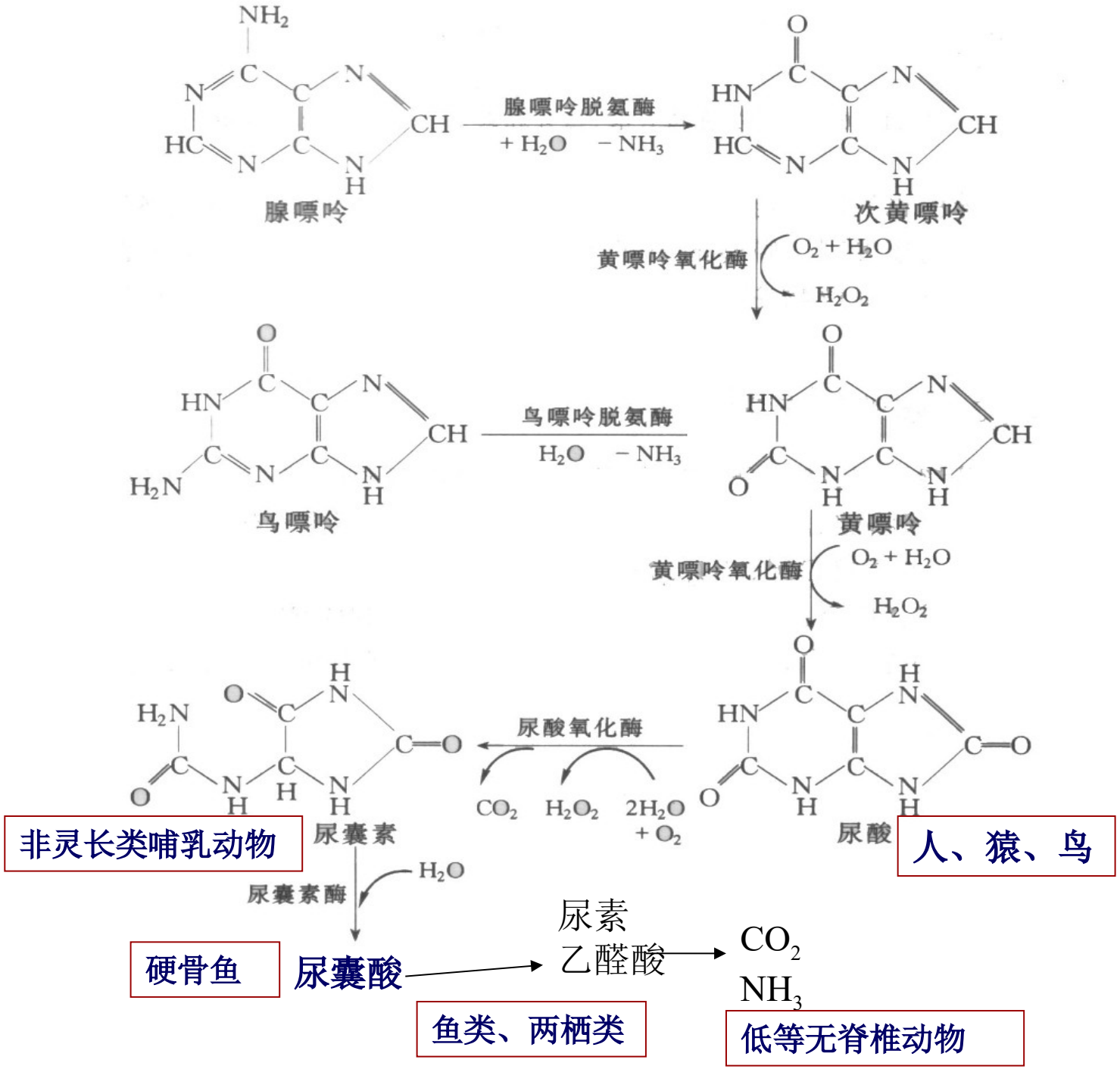
1、嘌呤的降解

2、嘧啶的降解

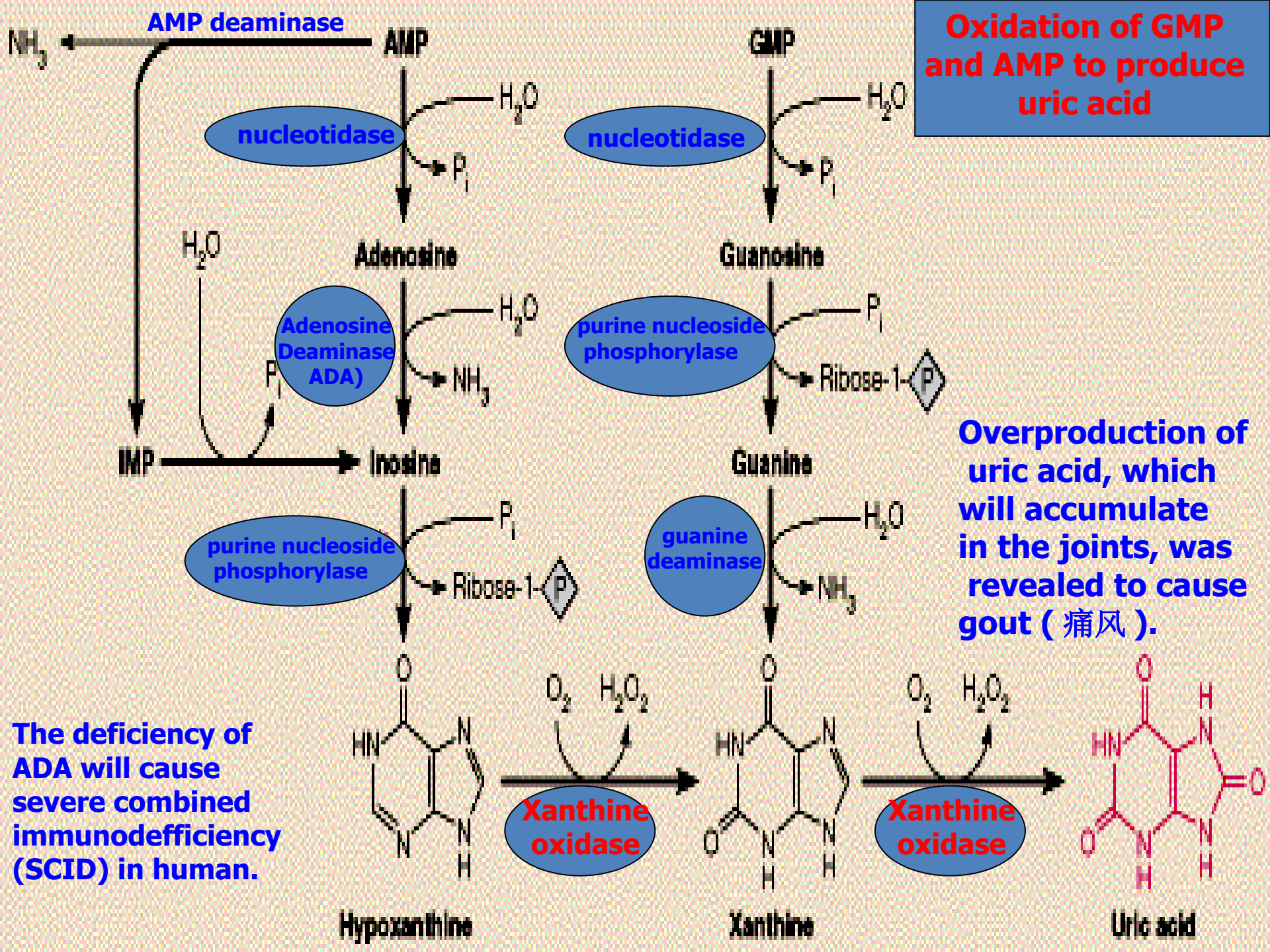
嘌呤的分解

- 腺嘌呤和鸟嘌呤要经过脱氨氧化转变为黄嘌呤再进行降解
- 不同生物分解嘌呤的酶系不一样
- 生物进化程度愈高，分解嘌呤的能力愈差




腺嘌呤
腺嘌呤脱氨酶
+ H2O − NH3
次黄嘌呤
黄嘌呤氧化酶
O2 + H2O
H2O2
鸟嘌呤
鸟嘌呤脱氨酶
H2O − NH3
黄嘌呤
黄嘌呤氧化酶
O2 + H2O
H2O2
尿酸
尿酸氧化酶
CO2 H2O2 2H2O + O2
尿囊素
尿囊素酶
H2O
尿囊酸
尿素
乙醛酸
CO2
NH3
非灵长类哺乳动物
人、猿、鸟
硬骨鱼
鱼类、两栖类
低等无脊椎动物

Oxidation of GMP and AMP to produce uric acid



Overproduction of uric acid, which will accumulate in the joints, was revealed to cause gout (痛风).

The deficiency of ADA will cause severe combined immunodefficiency (SCID) in human.

Uric acid is the excreted end product of purine nucleotide catabolism in humans and many other animals.



- 关键酶：黄嘌呤氧化酶
- 抑制剂—别嘌呤醇
- 最终产物：尿酸

正常人：血尿酸　　119-357 μmol/L （2-6 mg/dl）

男　　267.7 μmol/L （4.5 mg/dl）

女　　208.2 μmol/L （3.5 mg/dl）

血尿酸 ＞8 mg/dl

↓

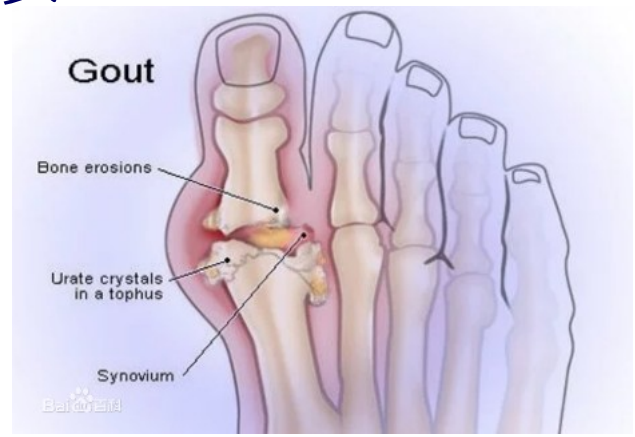
结晶

↓

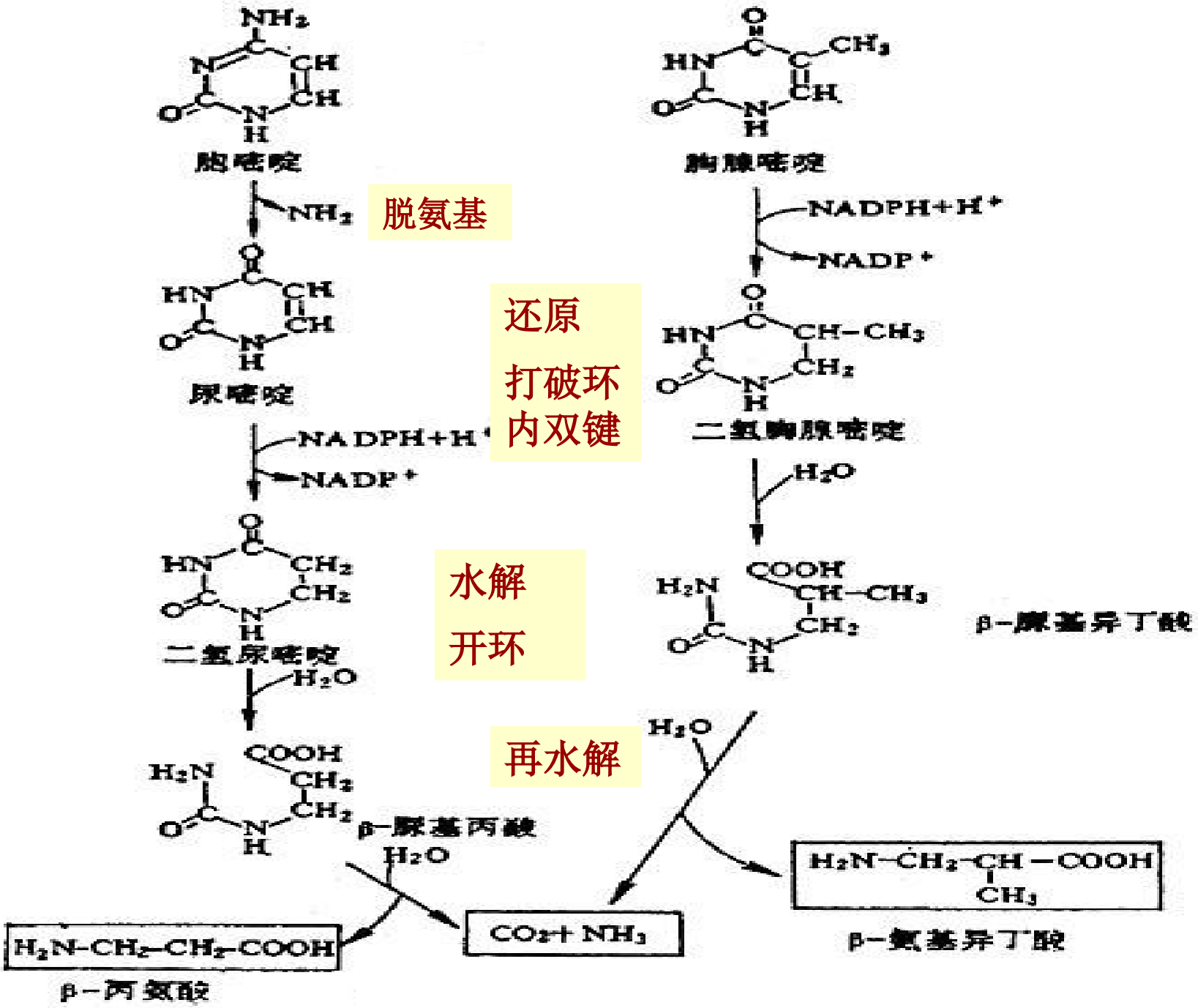
沉积组织

↓

痛风（gout）



嘧啶的分解


胞嘧啶
脱氨基
胸腺嘧啶
NADPH+H+
NADP+
还原
打破环内双键
尿嘧啶
二氢胸腺嘧啶
NADPH+H+
NADP+
H2O
水解
开环
二氢尿嘧啶
β-脲基异丁酸
H2O
再水解
β-脲基丙酸
H2O
H2N-CH2-CH2-COOH
β-丙氨酸
CO2+NH3
H2N-CH2-CH-COOH
CH3
β-氨基异丁酸

END

Thank you



A Note on the Nature of Science

In this twenty-first century, a typical science education often leaves the philosophical underpinnings of science unstated, or relies on oversimplified definitions. As you contemplate a career in science, it may be useful to consider once again the terms **science**, **scientist**, and **scientific method**.

Science is both a way of thinking about the natural world and the sum of the information and theory that result from such thinking. The power and success of science flow directly from its reliance on ideas that can be tested: information on natural phenomena that can be observed, measured, and reproduced and theories that have predictive value. The progress of science rests on a foundational assumption that is often unstated but crucial to the enterprise: that the laws governing forces and phenomena existing in the universe are not subject to change. The Nobel laureate Jacques Monod referred to this underlying assumption as the "postulate of objectivity." The natural world can therefore be understood by applying a process of inquiry—the scientific method. Science could not succeed in a universe that played tricks on us. Other than the postulate of objectivity, science makes no inviolate assumptions about the natural world. A useful scientific idea is one that (1) has been or can be reproducibly substantiated, (2) can be used to accurately predict new phenomena, and (3) focuses on the natural world or universe.



Science is both a way of thinking about the natural world and the sum of the information and theory that result from such thinking.

The power and success of science flow directly from its reliance on ideas that can be tested: information on natural phenomena that can be observed, measured, and reproduced and theories that have predictive value.



Σχιεντιστσ αρε ινδιωιδυαλσ ωηο ριγορουσλψ αππλψ
τηε σχιεντιφιχ μετηοδ το υνδερστανδ τηε νατυραλ ωορλδ.

Merely having an advanced degree in a scientific discipline does not make one a scientist, nor does the lack of such a degree prevent one from making important scientific contributions.

A scientist must be willing to **challenge any idea when new findings demand it.**



Τηε **σχιεντιφιχ μετηοδ** ισ αχτυαλλψ α χολλεχτιον οφ πατησ, αλλ οφ ωηιχη μαψ λεαδ το σχιεντιφιχ δισχοωερψ.

Ιν τηε ***hypothesis and experiment*** πατη, α σχιεντιστ ποσεσ α ηψποτηεσισ, τηεν συβφεχτσ ιτ το εξπεριμενταλ τεστ.

An analog of hypothesis and experiment is ***hypothesis and deduction.***

Crick reasoned that there must be an adaptor molecule that facilitated translation of the information in messenger RNA into protein. This adaptor hypothesis led to the discovery of transfer RNA by Mahlon Hoagland and Paul Zamecnik.

